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Experimental investigation on addition of furfuryl alcohol to diesel plastic fuel blends and optimization using Kissing Numbers

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Managing plastic waste while ensuring sustainable energy solutions is one of the critical challenges in modern engineering. The present study aims to address the combined issues of plastic waste management and engine exhaust gas emissions by replacing 50% of diesel with 50% of oil produced from plastic waste. Experimental tests were conducted on a common rail direct injection engine using Diesel Plastic Fuel (DPF = 50% Diesel + 50% Waste Plastic Fuel) and DPF blends containing 5%, 10%, and 15% Furfuryl Alcohol (FA), such as DPF-FA5, DPF-FA10, and DPF-FA15 across engine loads ranging from 20% to 100%. Key parameters such as Brake Thermal Efficiency (BTE), Brake Specific Fuel Consumption (BSFC), cylinder pressure, heat release rate, and emissions (CO, CO_2, HC, NO_x, and smoke opacity) were analyzed. The results were normalized and integrated into a Python-based program to estimate the performance index, adapting a novel optimization method using Kissing Numbers that allows the identification of optimal fuel configurations. The addition of FA improved engine combustion which resulted in an 8.2% increase in BTE, a 9.8% reduction in BSFC, and a 23.4% decrease in smoke opacity compared to DPF. While NO_x emissions increased by 19.5%, CO and HC emissions were reduced by 2.9% and 13.1%, respectively. DPF-FA15 was identified as the optimal blend for high loads that demonstrate the potential of FA to balance performance and emission characteristics using Kissing Numbers. The results highlight the potential of FA as a sustainable additive for diesel-plastic blends that offer a viable solution to reduce plastic waste and enhance engine performance and environmental sustainability.

Keywords CRDi engine, Diesel plastic fuel, Furfuryl alcohol, Kissing Numbers, Optimization, Waste plastic fuel

Abbreviations

ANN	Artificial neural networks
BSFC	Brake specific fuel consumption
BTE	Brake thermal efficiency
CHRR	Cumulative heat release rate
CO	Carbon monoxide
CO_2	Carbon dioxide
CRDi	Common rail direct injection
DPF	Diesel Plastic Fuel (50% diesel + 50% WPF)
DPF-FA5	95% DPF + 5% FA
DPF-FA10	90% DPF + 10% FA
DPF-FA15	85% DPF + 15% FA
ECU	Electronic control unit
EGT	Exhaust gas temperature
FA	Furfuryl alcohol
FTIR	Fourier transform infrared spectroscopy
GA	Genetic algorithms

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GC–MS	Gas chromatography/ Mass spectrometry
GRA	Grey relational analysis
GRG	Grey relational grade
HC	Hydrocarbon
HDPE	High-density polyethylene
HRR	Heat release rate
LDPE	Low-density polyethylene
NOx	Oxides of nitrogen
PI	Performance index
PM	Particulate matter
RSM	Response surface methodology
SOC	Start of combustion
WPF	Waste plastic fuel
SS	Stainless Steel

Symbols

U_c	Individual uncertainties
U_{x_i}	Uncertainty of the *i*th measured parameter
U_r	Uncertainty due to repeatability
S	Standard deviation
U_y	Overall uncertainty
K_n	Kissing Numbers
W_1, W_2 and W_3	Weights of each parameter
w_k	The normalized weight of the factor
$\Delta_{\min}$ and $\Delta_{\max}$	The minimum and maximum differences among all experimental values
Δ_{ij}	The absolute difference between the reference value and the normalized experimental value
ζ	The coefficient to maintain balance of weights

Reliable and well-integrated transportation networks are essential for a nation's economic advancement. Transportation is essential to contemporary life, as individuals depend on it to travel to and from diverse locations[1,2]. These benefits are inherently connected to the dangers that various means of mobility, particularly automobiles, present to the environment and human health[3–5]. The decline in pollution levels has grave repercussions for both humans and the ecosystem, with vehicular emissions being a major factor in outdoor air pollution[4,6–9]. Governments are attempting to mitigate transportation emissions by enforcing more stringent laws[10–12]. Mitigating the severity of exhaust emissions is essential by improving technical specifications via innovations, routine inspections, and utilizing cleaner fuels. An approach for effectively reducing individual emissions is transitioning to cleaner fuels[13–15]. Consequently, researchers explore innovative methods to enhance fuel efficiency while maintaining engine power by adapting alternative fuels reconciling mobility requirements with ecological preservation[16–19].

Due to their unmatched usefulness and extensive applications, plastics have become essential in modern-day life. Plastics are produced from petroleum-derived long hydrocarbon chains[20]. Approximately 4% of worldwide oil and gas production serves as feedstock for plastics, while 3–4% is utilized to supply energy for their production[21]. The research on plastic waste management indicated that merely 9% of global plastic is recycled[22]. Plastics are non-biodegradable substances with different chemical additives, including colouring agents, antioxidants, and stabilizers, have a pronounced tendency to leach into and contaminate the surrounding environment upon disposal due to photodegradation. The environmental implications of plastic waste are evident in the increasing levels of worldwide plastic contamination on both water and land. Plastic requires billions of years for natural biodegradation[23]. Improper disposal of plastic waste leads to further issues, including drain obstructions and health complications for humans. Disposing of plastic waste not only damages the environment but also represents lost potential for economic progress[24–26]. Moreover, recycling plastic for reuse conserves energy relative to the production of virgin plastic[27]. Researchers are exploring novel methods to tackle the combined challenges of plastic waste management along with fossil fuel dependency, including the conversion of plastic waste into usable alternative fuels.

Plastic waste can be treated through mechanical, catalytic, and thermal conversion, which successfully minimizes waste pollution and reduces reliance on fossil fuels for various purposes[28,29]. Research has established that Waste Plastic Fuel (WPF) possesses qualities analogous to those of diesel fuel[24,30–33]. Research demonstrates that engines utilizing pure WPF typically demonstrate poor performance relative to diesel[34–37]. The brake thermal efficiency (BTE) of engines powered by WPF is often reduced due to issues including combustion delay and higher fuel consumption, attributed to WPF's reduced cetane number and high viscosity[38–40]. The chemical kinetics of WPF indicate that the thermal cracking influences both long-chain and short-chain carbon molecules, leading to the development of lean mixtures as well as lean combustion during the delay phase[41–43]. Regarding emissions, neat WPF often generates more amount of pollutants than diesel. Carbon monoxide (CO) along with hydrocarbon (HC) emissions are increased, mostly because of incomplete fuel combustion[44–46]. Moreover, WPF typically produces increased oxides of nitrogen (NO_x) as well as Particulate matter (PM) emissions because of its elevated aromatic content and higher burning temperatures[34,45,47,48]. These emissions present a substantial environmental concern, highlighting the necessity for after-treatment equipment like catalytic converters and particulate filters to alleviate their effects. WPF consists of long polymer chains that require a specific amount of energy to disrupt. Nevertheless, despite these disadvantages, the blending of WPF with diesel has demonstrated potential in comparing performance and emissions. While maintaining engine performance, lower blend ratios

such as 10–30% WPF combined with diesel have shown better combustion stability and lower emissions as compared to pure WPF[24,49]. Nevertheless, as the concentration of WPF in the mixture increases, the detrimental impacts on engine performance and emissions become increasingly evident. Extending upon this basis, the emphasis shifts to mitigating these challenges through the investigation of oxygenated additives strategies to improve the performance and sustainability of WPF blends.

Oxygenated additives exhibit low Heat Release Rate (HRR) during the initial phase of cool flame formation but demonstrate normal to high heat release rate throughout the mid-stage of blue flame oxidation, followed by a standard red flame that maintains a consistent heat output during complete combustion. The strategic addition of alcoholic additives into WPF diesel blends has proven to be an effective method for improving engine performance and addressing the environmental issues related to WPF[50,51]. Alcohols like ethanol[50], methanol[52], n-butanol[53], isopropanol[51], and n-propyl alcohol[54] have distinctive physicochemical properties, including elevated oxygen content, reduced viscosity, and increased volatility, which might beneficially affect combustion characteristics as well as emissions in diesel engines[55]. Research indicates that adding 10–20% alcohols into WPF-diesel blends improves combustion and diminishes ignition delays, offsetting the reduced cetane number and increased viscosity of WPF[56]. Alcohols additionally aid in reducing combustion temperatures, encountered with pure WPF or rich blends of WPF[57]. Furthermore, the use of alcohols enhances the miscibility of WPF and diesel, ensures consistent combustion, and diminishes the probability of phase separation, a prevalent problem with WPF diesel systems. Alcohols facilitate cleaner combustion by supplying an inherent oxygen source, hence accelerating oxidation reactions throughout the combustion process. Research indicates significant decrease in CO and HC emissions resulting from enhanced combustion efficiency[58,59]. PM emissions are diminished as the oxygenated characteristics of alcohols slow down the generation of soot precursors[34]. Nonetheless, there exists a compromise with NO_x emissions. Although alcohols often reduce peak combustion temperatures, which may minimize thermal NO_x generation, their elevated oxygen concentration can nevertheless result in increased NO_x under some conditions[60]. Building on this foundation, the focus switches to optimization techniques adapted for increased sustainability.

Optimization is crucial for improving the performance, efficiency, and emissions of diesel engines, particularly when utilizing alternative fuels such as WPF-diesel blends. The intricate relationship among fuel characteristics, combustion behavior, and engine operating parameters requires the implementation of optimization methods to attain optimal performance[17]. Researchers have utilized many optimization techniques, encompassing experimental methods such as Response surface methodology (RSM)[31,40], Artificial neural networks (ANN)[44], and Genetic algorithms (GA) apart from hybrid approaches that integrate computational and experimental strategies[54]. These methodologies facilitate systematic investigation of parameter interactions, including injection timing, blend ratios, and compression ratios, to determine optimal operating conditions[58]. Every optimization technique presents unique benefits. For example, RSM effectively models non-linear interactions and enables multi-objective optimization, whereas ANN focuses on forecasting outcomes from intricate datasets. GA is very effective at addressing multi-variable, non-linear optimization challenges. Nonetheless, these methodologies possess constraints, such as computational intensity (ANN and GA) and dependence on precise experimental data for model calibration (RSM). Despite these challenges, optimization is essential for leveraging the possibilities of fuel blends ensuring that engines attain enhanced efficiency and decreased emissions without sacrificing operational stability.

The novelty of this study lies in the synergistic integration of a high-percentage (50%) waste plastic fuel (WPF) blend with diesel and the introduction of Furfuryl Alcohol (FA) as a novel oxygenated additive—an area not previously explored in diesel-WPF combustion research[13,30,31,45,53,57]. Unlike commonly studied alcohols such as ethanol, methanol, and butanol, FA offers unique chemical properties due to its furan ring structure, which facilitates early-stage oxidation and stable combustion. This enhances atomization, reduces ignition delay, and promotes efficient burning of heavy hydrocarbons present in WPF. Additionally, the study introduces an innovative optimization methodology based on the concept of Kissing Numbers to identify the optimal FA concentration for balancing engine performance and emissions. This dual approach—leveraging an underexplored additive (FA) and a novel multi-objective optimization strategy—presents a significant advancement in the development of sustainable fuel alternatives, offering an effective solution for plastic waste valorization while enhancing combustion efficiency and reducing harmful emissions in diesel engines.

The aim of the present study is to address the combined issues of plastic waste management along with engine exhaust gas emissions by replacing 50% of diesel with 50% of oil produced from plastic waste and evaluate the influence of FA on engine performance, combustion behavior, and exhaust emissions in a Common Rail Direct Injection (CRDi) engine under varying load conditions. In addition, a novel optimization methodology based on Kissing Numbers is employed to identify the optimal FA concentration that ensure a balanced trade-off between performance and emissions that include brake thermal efficiency (BTE), brake-specific fuel consumption (BSFC), and pollutant emissions such as NO_x, CO, HC, and smoke opacity. By integrating experimental analysis with a novel optimization approach, this research seeks to establish a viable and environmentally sustainable fuel strategy for diesel engines that contribute to cleaner combustion technologies and efficient plastic waste utilization.

Materials and methods

Waste plastic fuel production

The pyrolysis apparatus employed in this experiment is a batch reactor, including a Stainless Steel (SS) tube sealed at one end and an outlet tube at the opposite end for the collection of reaction products. The SS tube is externally heated by an electric furnace, with temperature monitored by a chromium-aluminum K-type thermocouple affixed within the reactor, while temperature regulation is managed by an external PID controller at an accuracy of ± 3.6°C. Each pyrolysis reaction involved the loading of 30 g of plastic materials, including High-Density

Polyethylene (HDPE) and Low-Density Polyethylene (LDPE). The proximate and ultimate analysis of plastic waste is presented in Table 1. A combination of catalyst and plastic samples in varying catalyst-to-plastic ratios was subjected to pyrolysis, heated at a rate of 20 °C/min until the specified temperature was reached. The condensable liquid products were collected via the condenser and measured. The solid residue remaining in the reactor post-pyrolysis was weighed. The weight of the gaseous product was determined by material balance calculations. Pyrolysis was conducted at various temperatures between 400 and 550 °C. The detailed production of plastic fuel is presented in Pappula et al.[61].

Fuel blend preparation

The fuel blends used in the present investigation were prepared with precision to ensure uniformity and adaptability for application in a diesel engine. The test fuels were neat diesel, neat WPF, DPF (50% diesel + 50% WPF), DPF-FA5 (95% DPF + 5% FA), DPF-FA10 (90% DPF + 10% FA), and DPF-FA15 (85% DPF + 15% FA). The DPF was chosen as the base fuel blend to meet the 50% reduction in diesel consumption required, that aligns with global efforts to reduce dependency on fossil fuels. The effects of Lower WPF blends (10–30%) have been well documented in scientific literature[21,24,31,34], but the impact on diesel replacement is minimal. However, high WPO blends (≥ 50%) exhibits concern such as higher ignition delay, combustion instability, and increased emissions that make them less viable for CRDi engine operation, while this deficiency is substituted by the addition of FA. High oxygen content (32.62%) in FA facilitates better fuel oxidation that improves combustion efficiency. Lower viscosity than WPO in FA improves fuel atomization and reduces incomplete combustion. The FA concentrations of 5%, 10%, and 15% were selected to systematically investigate the effect of incremental oxygen enrichment on combustion and emissions. High-speed diesel available on the market was acquired and utilized as the principal reference fuel. WPF, produced via the pyrolysis of plastic waste, was purified to eliminate contaminants and attain a grade appropriate for diesel mixing. The mixing was done in a stirring chamber to ensure homogeneity and avoid phase separation. Required quantities of diesel and WPF were initially measured using precision burettes and subsequently transferred into a calibrated blending chamber. The mixture was agitated mechanically at a constant speed for 15 min to ensure perfect miscibility of diesel and WPF. For the DPF-FA blends, FA was blended into the pre-mixed DPF at the specified ratios. The mixture was agitated for an extra 10 min to attain a uniform blend. The mixed fuels were permitted to remain for 24 h to assess phase stability and were visually examined for indications of separation[44]. All fuel mixtures were preserved in airtight containers to avert contamination and evaporation losses. The containers were maintained in a temperature-regulated environment to assure stability and prevent oxidation, especially for blends containing FA, which is susceptible to hygroscopic effects[38]. The summary of principal characteristics of the test fuels, as estimated using ASTM standards[62] is presented in Table 2.

Engine test rig and test procedure

The present study utilizes a Kirloskar TV1 diesel engine, a single-cylinder CRDi engine with water cooling, operating at 1500 rpm, 600 bar fuel injection pressure, and producing 3.5 kW of power to examine the effects of various factors on engine performance, combustion and emissions. The Electronic Control Unit (ECU) is coupled with the pressure sensor. Loading is accomplished via an eddy current dynamometer. Temperature readings are acquired utilizing K-type thermocouples at several locations, encompassing air intake, fuel intake, cooling system, and exhaust gas. Table 3 presents the technical specifications of the engine. The study intends to evaluate the engine parameters at a constant speed of 1500 rpm under varied load conditions of 20%, 40%, 60%, 80%, and 100% using test fuels including diesel, WPO, DPF, DPF-FA5, DPF-FA10, and DPF-FA15. The engine was primarily operated on diesel fuel without additives to gather baseline data. The experiment involves quantifying several characteristics during combustion, such as in-cylinder pressure, HRR, and ignition delay. Performance characteristics such as BTE and Brake specific fuel consumption (BSFC) were calculated from the engine output data. Exhaust emissions including NO_x, CO_2, CO, HC, and smoke opacity are assessed utilizing a five-gas analyzer (AVL Digas 444N) as well as a smoke meter (AVL 437C), respectively. The pertinent parameters are captured by a data logger, and a computer is installed with "Engine Soft" software. To ensure

S. no	Property	Value
Proximate analysis		
1	Volatile matter (wt.%)	95.67
2	Ash content (wt.%)	1.75
3	Fixed carbon (wt.%)	1.92
4	Moisture content (wt.%)	0.66
Ultimate analysis		
S. No	Property	Value
1	Carbon (wt.%)	79.84
2	Hydrogen (wt.%)	16.32
3	Oxygen (wt.%)	3.74
4	Nitrogen (wt.%)	0.09
5	Sulphur (wt.%)	0.01

Table 1. Proximate and ultimate analysis of plastic waste.

S. no	Property	Diesel	WPF	DPF	FA	DPF-FA5	DPF-FA10	DPF-FA15	Instrument used
1	Density (kg/m^3)	840 ± 0.5	845 ± 0.5	842.5 ± 0.5	1129 ± 0.5	856.8 ± 0.5	871.2 ± 0.5	885.5 ± 0.5	SVM 3001 Cold Properties; Anton Paar
2	Kinematic viscosity cST @40 °C	2.12 ± 0.02	3.64 ± 0.03	2.88 ± 0.02	4.16 ± 0.05	2.94 ± 0.02	3.01 ± 0.03	3.07 ± 0.03	SVM 3001 Cold Properties; Anton Paar
3	Calorific value (kJ/kg)	46,500 ± 150	42,900 ± 170	44,700 ± 160	22,000 ± 200	43,565 ± 170	42,430 ± 175	41,295 ± 180	e2K oxygen bomb calorimeter
4	Cetane number	54.5	44	49	10.8	47.09	45.18	43.27	Two variable cetane index calculation
5	Flash point (°C)	54 ± 0.5	62 ± 0.7	58 ± 0.6	65 ± 1.0	58 ± 0.6	59 ± 0.7	59 ± 0.8	Automatic flash point tester, Pensky Martens, Scavini
6	Self-ignition temperature (°C)	210 ± 2.0	258 ± 3.0	242 ± 2.5	490 ± 4.0	254 ± 2.5	267 ± 2.8	279 ± 0.8	Glass flask heated in controlled environment
7	Carbon content (wt.%)	85 ± 0.2	780.3	82 ± 0.25	61.21 ± 0.5	81 ± 0.25	80 ± 0.3	79 ± 0.35	FLASH 2000 CHNS/O Analyzer, Thermo Scientific
8	Nitrogen content (wt.%)	0 ± 0.0	1.29 ± 0.02	0.91 ± 0.02	0 ± 0.0	0.86 ± 0.01	0.82 ± 0.01	0.77 ± 0.01	FLASH 2000 CHNS/O Analyzer, Thermo Scientific
9	Hydrogen content (wt.%)	13 ± 0.1	12 ± 0.15	12 ± 0.12	6.17 ± 0.2	12 ± 0.12	11 ± 0.15	11 ± 0.18	FLASH 2000 CHNS/O Analyzer, Thermo Scientific
10	Oxygen content (wt.%)	0 ± 0.0	0 ± 0.0	0 ± 0.0	32.62 ± 0.3	1.63 ± 0.05	3.26 ± 0.07	4.89 ± 0.1	FLASH 2000 CHNS/O Analyzer, Thermo Scientific

Table 2. Properties of base fuels and blends.

S. no	Parameter	Remarks
1	No. of cylinders	01
2	Bore (mm)	87.5
3	Stroke (mm)	110
4	Swept volume (cc)	661
5	Rated speed (rpm)	1500
6	Rated power @1500 rpm (kW)	3.5
7	Fuel injection timing (°CA) bTDC	23
8	Fuel injection pressure (bar)	600
9	Engine cooling	Water

Table 3. The specifications of CRDi engine.

precision, the engine runs for approximately 15 min, followed by a stabilization period of 10 to 15 min before commencing the experiment. Furthermore, to enhance accuracy, the mean results from three measurements are considered[56]. The schematic of the experimental setup is shown in Fig. 1 and the measuring range, resolution and accuracy of the instruments used is detailed in Table 4.

Uncertainty analysis

Uncertainty analysis is crucial in experimental studies to assess the reliability of measured results. The uncertainties arise from measurement instruments, calibration, environmental factors, and test procedures[45]. The present study evaluates combined uncertainty U_c for the measured parameters is calculated, using the Eq. (1)[56], as the root-sum-square of individual uncertainties (U_c).

$$U_c = \sqrt{\sum_{i=1}^{n} U_{x_i}^2} \tag{1}$$

where U_{x_i} represents the uncertainty of the ith measured parameter, and n is the total number of measured parameters. The uncertainty due to repeatability (U_r) is determined using the standard deviation (S) of repeated measurements as shown in Eq. (2)[35].

$$U_r = \frac{S}{\sqrt{n}} \tag{2}$$

where n is the number of repeated trials, three in the present study. The total uncertainty (U_T) for a given parameter is the sum of instrumental and repeatability uncertainties, as shown in Eq. (3).

$$U_r = \sqrt{{U_c}^2 + {U_r}^2} \tag{3}$$

For parameters derived from multiple measurements such as BTE or BSFC, the overall uncertainty (U_y) is estimated using Eq. (4)[63].

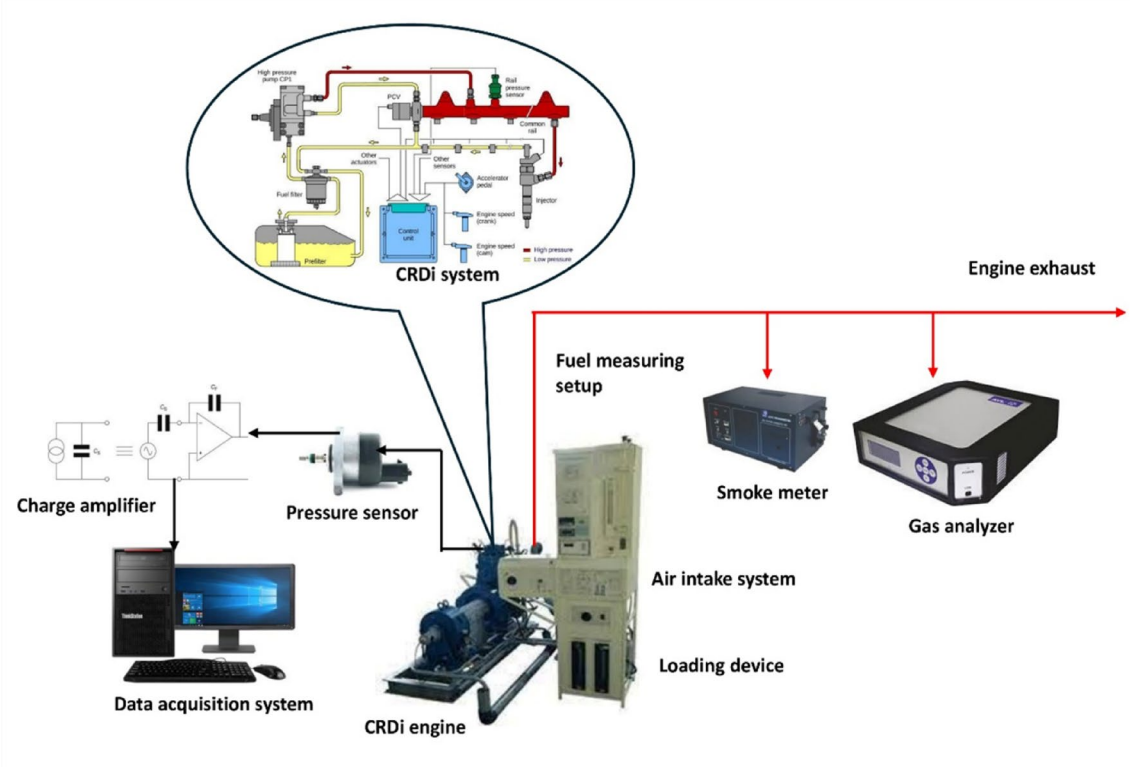


Fig. 1. The schematic of experimental setup.

S. no	Measured quantity	Range	Resolution	Accuracy
1	CO	0–15%	0.001%	± 0.02%
2	CO_2	0–20%	± 0.1%	± 0.03%
3	HC	0–20,000 ppm	1 ppm	± 10 ppm
4	NO_x	0–6000 ppm	1 ppm	± 5 ppm
5	Smoke opacity	0–100%	0.1%	± 1%
6	Speed	0–2000 rpm	5 rpm	± 10 rpm
7	EGT	0–1000 °C	± 1 °C	± 1 °C

Table 4. The measuring range, resolution and accuracy of the instruments used.

$$U_y = \sqrt{\left(\frac{\partial y}{\partial x_1} U_{x_1}\right)^2 + \left(\frac{\partial y}{\partial x_2} U_{x_2}\right)^2} + \cdots \tag{4}$$

where y is the derived metric, $x_1, x_2, \ldots$, are the individual parameters, and $U_{x_1}, U_{x_2}, \ldots$, are their uncertainties. In the present study, the overall uncertainty is ± 3.11%.

Optimization using Kissing Numbers

The optimization techniques are important to identify the optimal fuel blend configuration from the perspective of engine performance and emission characteristics. Methods like RSM, GA, and ANN are widely used in multi-objective optimization. But these methods have some limitations, for example, RSM is a statistical technique that requires a predefined polynomial model that may not capture all the possible interactions between fuel properties and combustion parameters. GA and ANN are computationally expensive as they depend on large datasets and iterative training, respectively, and are therefore infeasible for application to experimental studies with limited data. Optimization using Kissing Numbers is a heuristic method inspired by the mathematical concept of "Kissing Numbers" (K_n) that represents the maximum number of non-overlapping spheres in R^n(n is dimensional Euclidean space) that can simultaneously touch another sphere of the same size[64]. In simpler terms, the K_n represents the maximum number of neighboring entities that can tangentially "kiss" a central entity in each dimensional space. Optimization using K_n uses direct geometric evaluation to identify optimal solutions, unlike ANN and GA. It also allows simultaneous optimization of performance and emissions characteristics

without requiring extensive computational resources. In addition, optimization using K_n provides a transparent ranking of fuel blends that make it easier to validate, unlike black-box models in ANN.

Kissing Number

The K_n problem has intrigued mathematicians for centuries in the case of three dimension (R^3). The problem can be visualized to find how many "white carrom coins" can simultaneously touch a "black carrom coin" on a carrom board without overlapping. The most symmetrical arrangement involves 12 white coins placed around the central black coin that corresponds to the vertices of a regular octahedron centered on the black coin. Despite this arrangement, the 12 outer coins do not touch each other and can move freely, which raises a hypothetical question: Could a 13th coin also fit if all 12 coins are shifted slightly? This question was famously debated between Isaac Newton and David Gregory in 1694. Newton's position was that the maximum Kissing Number in three dimensions (K_3) is 12. This is achieved when the outer coins are symmetrically placed, as in the octahedral arrangement. Whereas Gregory speculated that K_3 = 13, reasoning that the unit sphere's surface area (≈ 14.9) might accommodate more overlapping regions. This debate was later resolved mathematically by Coxeter and confirmed Newton's assertion that K_3 = 12. The complexity and subtleties of the problem underscore its relevance in understanding spatial packing and symmetry. The Kissing Number problem becomes increasingly complex as the dimensionality of the space increases[65,66]. Coxeter, in 1963, proposed upper bounds for K_n in dimensions $n = 4, 5, 6, 7,$ and 8, based on geometric considerations such as, $K_4 \leq 26$, $K_5 \leq 48$, $K_6 \leq 85$, $K_7 \leq 146$, and $K_8 \leq 244$. These bounds provide insights into the limits of sphere packing efficiency and serve as foundational knowledge for various applications in optimization, data clustering, as well as computational geometry[67–69]. The problem in R^2 and R^3 are shown in Fig. 2[64].

The principles of K_n are inspired by the algorithms and methodologies in modern optimization problems which include the use of K_n in the present study. By leveraging the spatial symmetry and packing efficiency captured in the Kissing Number method, the present study applies these principles to determine the optimal configurations of test fuel blends among DPF, DPF-FA5, DPF-FA10, and DPF-FA15 across different load conditions to maximize BTE, minimize BSFC and emissions. The optimization methodology using K_n in this study follows a structured approach:

- Step 1: Data normalization- Experimental data from engine tests (BTE, BSFC, NO_x, CO, CO_2, HC, Smoke opacity) are normalized to eliminate unit dependency.
- Step 2: Performance index calculation- A weighted Performance Index (PI) is calculated for each fuel blend across different engine loads. Weighting factors were assigned based on the relative significance of each parameter.
- Step 3: Application of K_n- The K_n principle is used to compare fuel blend performance at different loads. All fuel blends are then iteratively paired and evaluated using the maximum sphere packing principle to determine the optimal configuration.
- Step 4: Selection of optimal fuel blend- The blend with the highest PI value is identified as optimal for each load condition. The optimization constraints ensures that no redundant or unrealistic fuel configurations are selected.

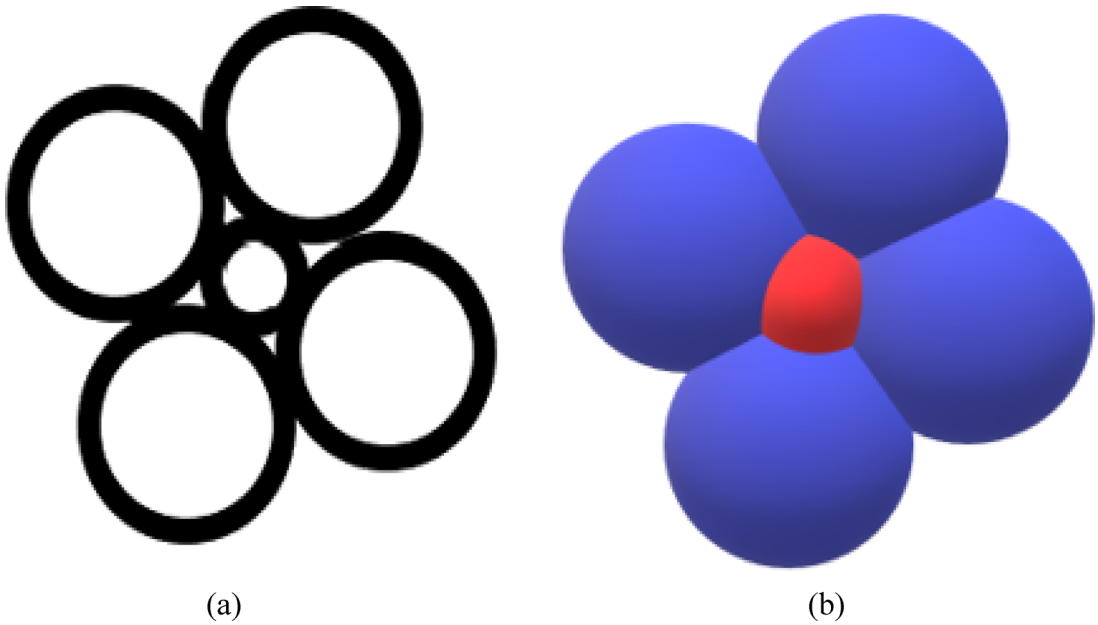


Fig. 2. The problem in (**a**) R^2, and (**b**) R^3.

Data normalization

Data normalization is important to ensure comparability and eliminate the influence of varying units and scales across performance and emission metrics. Normalization helps to put the parameters and the outcomes in a similar range for analysis and improvement of all the measured and estimated engine characteristics such as BTE, BSFC and emissions. The data normalization method adapted in this study is based on the procedure proposed by Narayanan Kannaiyan and Sekar[70]. "Higher the better" is beneficial for BTE and CO_2 as they indicate more efficient fuel energy conversion and complete combustion. "Lower the better" is beneficial for BSFC to improve fuel economy and for lower levels of emissions such as NO_x, CO, HC, and smoke opacity as they reflect reduced environmental impact and cleaner combustion. This differentiation ensures that each parameter is considered and normalized appropriately.

Performance index calculation

The Performance index (PI) is calculated by integrating multiple normalized metrics into a single score to evaluate and optimize the overall performance of the engine. Weights are assigned to key performance and emission parameters based on their relative significance to calculate the PI using a weighted sum approach[71]. This makes this method effective to ensure that the optimization process corresponds to the focus of improvement in efficiency, fuel consumption and emissions. PI is calculated using the Eq. (5).

$$PI = W_1 \times Normalized\ BTE - W_2 \times Normalized\ BSFC - W_3 \times Average\ normalized\ emission \quad (5)$$

here, W_1, W_2, and W_3 are the weights of each parameter, determined to balance their importance in achieving optimal engine performance.

Application of Kissing Numbers method

BTE represents the percentage of fuel energy converted into useful work. Higher BTE signifies enhanced engine efficiency, a crucial goal in fuel performance research. Consequently, BTE is assigned to have the greatest significance in the PI computation (W_1 = 0.4)[71]. BSFC measures the fuel consumed per unit of power output. Lower BSFC values reflect better fuel economy that is essential for sustainable engine operation. Nevertheless, BSFC is crucial, it is assigned slightly less weight than BTE to emphasize overall energy efficiency (W_2 = 0.3). Weight of emissions are averaged and assigned a collective weight (W_3 = 0.3). Reducing emissions is essential to mitigate environmental effect and adhere to emission regulations. The equal significance of BSFC and emissions demonstrates their mutual importance in sustainable engine performance. The average of the normalized values of these emissions assures that no individual pollutant disproportionately affects the overall emissions score.

Selection of optimal fuel blend

By evaluating configurations for their relative performance, the K_n method is used to further optimize the process. The method analyzes all pairs of configurations (i, j) to identify the combinations with the highest PI values for each load. Optimal configurations iterate through DPF, DPF-FA5, DPF-FA10, and DPF-FA15 for each load condition. A Python based program is developed to calculate K_n and PI. The flowchart for the optimization using K_n is presented in Fig. 3. The algorithm of the program is given in Appendix A.

Results and discussion

In-cylinder pressure

The quantity and quality of fuel utilized during the premixed combustion phase influences the in-cylinder pressure on the engine. Figure 4 illustrates the variation in in-cylinder pressure in relation to crank angles for all tested fuels under full load conditions. The peak in-cylinder pressure for diesel, WPF, DPF, DPF-FA5, DPF-FA10, and DPF-FA15 was found to be 104.60, 117.22, 106.12, 99.59, 94.25, and 91.39, respectively. The WPF curve exhibits the maximum peak pressure among all blends. This is due to its elevated aromatic content along with the delayed combustion, leading to a rapid pressure increase during the phase of late combustion[61]. Nonetheless, this may result in more severe operating conditions and possible engine durability concerns. Similar findings were observed in an investigation by Arani et al.[56], for neat WPF. On the contrary, Muthukumar and Kasiraman[72] observed a 5.93% decrease in peak in-cylinder pressure for neat WPF than diesel which was ascribed to the shorter ignition delay and poor premixed combustion. The maximum in-cylinder pressure for Diesel exceeds that of the DPF-FA blends, indicating enhanced combustion characteristics, such as its higher cetane number and ideal ignition characteristics. Diesel and DPF demonstrate an earlier increase in pressure relative to WPO as well as blends of DPF-FA. As the concentration of FA increases from DPF-FA5 to DPF-FA15, the peak in-cylinder pressure exhibits a slight decline. As compared with diesel, the peak in-cylinder pressure of DPF-FA, DPF-FA10, and DPF-FA15 decreased by 4.79%, 9.89%, and 12.62%, respectively. This is attributed to the lower calorific value as well as cetane number of FA, which decreases combustion intensity and delays the pressure rise.

The hydrocarbons in WPF experience thermal cracking, which decompose long-chain polymers into shorter alkanes and alkenes. FA, an oxygenated molecule, evaporates rapidly owing to its considerable volatility and thermally decomposes into smaller oxygenated intermediates, including furan derivatives and aldehydes, which facilitate primary combustion processes. During the ignition delay phase, FA promotes the generation of reactive oxygen species and free radicals, which catalyze the burning of hydrocarbons in DPF. The oxygenation impact of FA expedites low-temperature oxidation reactions, such as hydroperoxide breakdown, resulting in a more prompt and smoother transition to high-temperature combustion relative to neat DPF. The denser hydrocarbon fractions in the DPF experience diffusion-dominated combustion while fuel injection persists during the diffusion combustion phase.

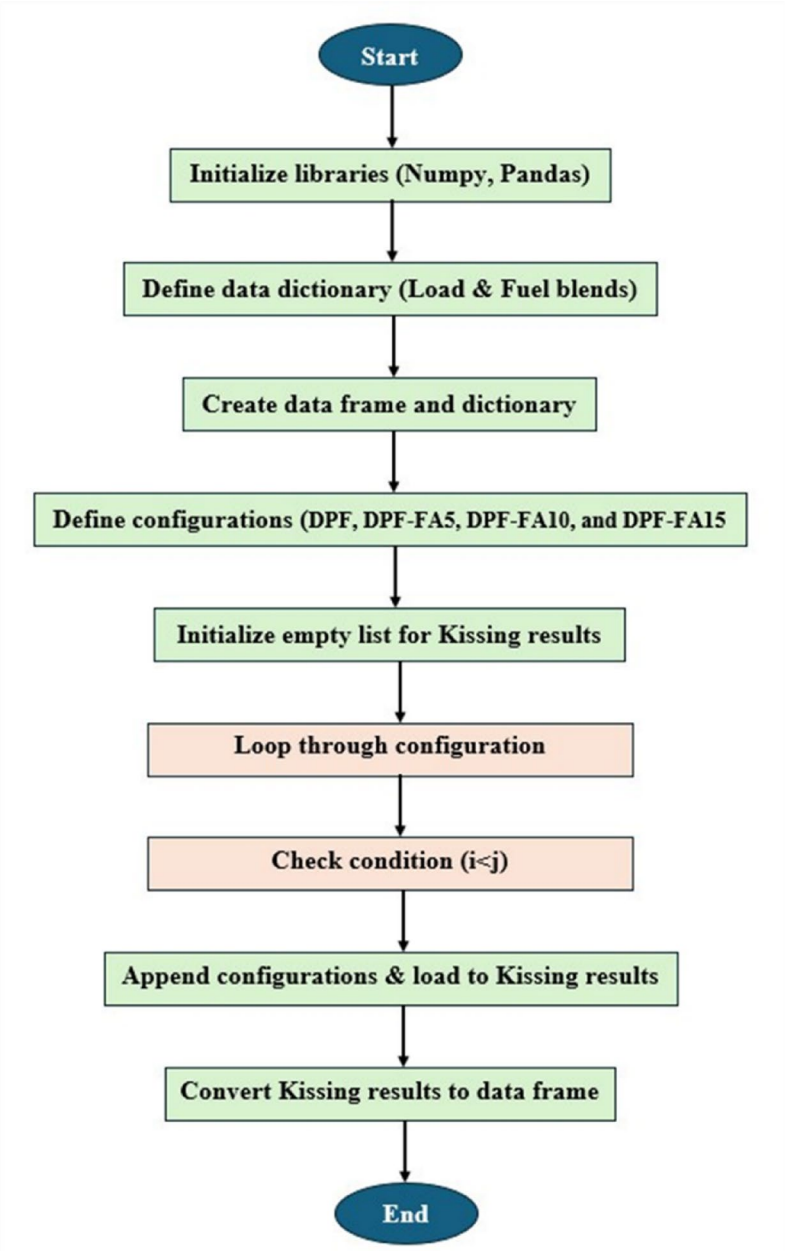


Fig. 3. The flowchart of the optimization using Kissing Numbers.

Net heat release rate

The HRR denotes the rate at which chemical energy is liberated by the fuel during combustion, measured from the cylinder pressure observations. Quantitative data regarding combustion progression can be extracted from HRR profiles. The HRR plots for all test fuels at 100% are presented in Fig. 5. The peak HRR for diesel, WPF, DPF, DPF-FA5, DPF-FA10, and DPF-FA15 was found to be 110.00, 118.22, 114.21, 98.92, 92.56, and 88.32 kJ/m^3deg, respectively. WPF exhibits a delayed, yet higher HRR in comparison to Diesel, achieving the highest peak as compared to all blends of fuel. The highest peak HRR in WPF is due to building up of unburnt fuel during an extended ignition delay, slower atomization, higher viscosity, resulting in a rapid energy release during the premixed period[24]. The DPF exhibits a somewhat higher HRR peak than Diesel, indicating a compromise in combustion quality attributable to the addition of WPF. The diminished heat content and inadequate mixing of air with fuel in diffusion combustion may account for the reduction in heat emission[61]. The HRR of Diesel is inferior to that of WPF and DPF due to its higher volatility, along with optimized fuel air mixing, which facilitate rapid combustion in the premixed phase[34]. Diesel exhibits the shortest ignition delay, as demonstrated by the earliest start of the HRR plot. Its enhanced ignition characteristics ensure rapid reactions of fuel–air after injection[73]. With the incremental addition of furfuryl alcohol in DPF-FA5, DPF-FA10, DPF-FA15, the peak HRR lowers consistently. The calorific value of DPF-FA blends regulates combustion intensity by enhancing oxygen availability, resulting in a more regulated heat release.

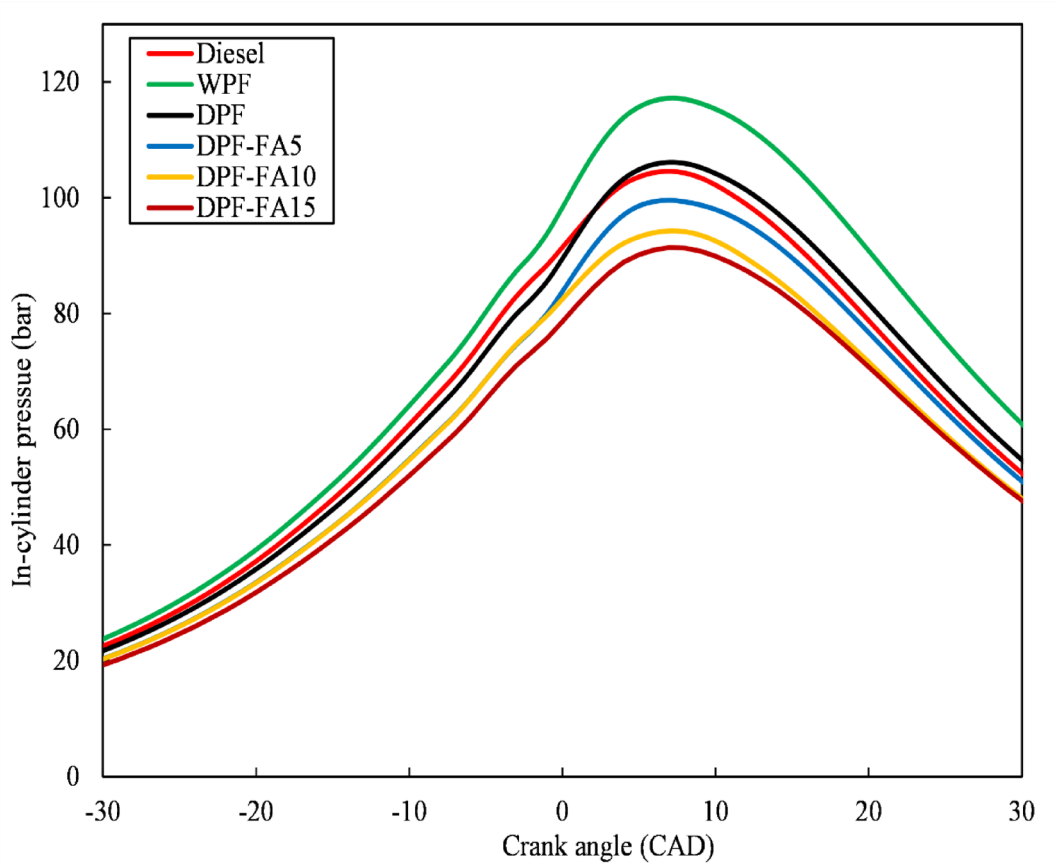


Fig. 4. The in-cylinder pressure variations of test fuels at 100% load.

Blends of DPF-FA promote accelerated low-temperature oxidation reactions during the ignition delay period owing to the oxygen content in FA. These reactions yield intermediate molecules such as hydroperoxides, which breakdown to produce radicals and ignite combustion. During the premixed combustion phase, the fast oxidation of volatile intermediates leads to a significant increase in HRR. The oxygenated composition of FA enhances HC oxidation, resulting in increased energy release and a more significant HRR peak relative to pure DPF. After the initial premixed phase, combustion shifts to a diffusion-controlled phase. During this phase, the denser HC fractions in the DPF persist in oxidizing, facilitated by the oxygen provided by FA. The oxygenation impact of FA minimizes the formation of unburned HC in the fuel, facilitating more thorough combustion during the diffusion phase. This facilitates prolonged HRR across an extended crank angle duration. At low FA concentrations, the HRR profile exhibits a more pronounced peak attributable to the strengthening of the premixed combustion phase. This signifies effective energy discharge and enhanced combustion dynamics. At elevated FA concentrations, in the case of DPF-FA10 and DPF-FA15, the HRR profile broadens, indicating a more regulated and stable combustion process. The oxygenated characteristics of FA diminish the intensity of combustion, resulting in more gradual transitions between combustion phases.

Cumulative heat release rate

The Cumulative Heat Release Rate (CHRR) profiles, as shown in Fig. 6, for the test fuels presents the influence of fuel chemistries on the combustion process. Although the combustion chamber geometry and operational conditions remained unchanged, distinct variations in energy release were observed, driven by the type and composition of the fuels. The start of combustion (SOC) occurs earlier for the FA-blended fuels than for conventional diesel and WPF. DPF-FA15 initiates combustion around -2° CAD and diesel combustion begins slightly later. This advancement in SOC can be attributed to the presence of oxygenated compounds in FA that facilitate quicker low temperature reactions and shorten the ignition delay period.

Fuel blends with FA additive tend to complete combustion over a shorter crank angle range, with DPF-FA15 exhibiting the shorter combustion duration (around 37° CAD). This is an encouraging indicator of efficient combustion, suggesting that a larger fraction of the energy is released during the optimal portion of the expansion stroke[35]. The CHRR slope is noticeably steeper in FA-based blends that reflects the rate of energy release. The heat accumulation rate between -1° and + 10° CAD shows a more intense premixed combustion phase for blends of FA additive. DPF and WPF fuels show a more gradual rate of energy release that reflects slower and less intense combustion due to their relatively lower volatility and absence of intrinsic oxygen[52]. Considering the location of peak combustion, FA additive fuel blends reach their maximum cumulative heat

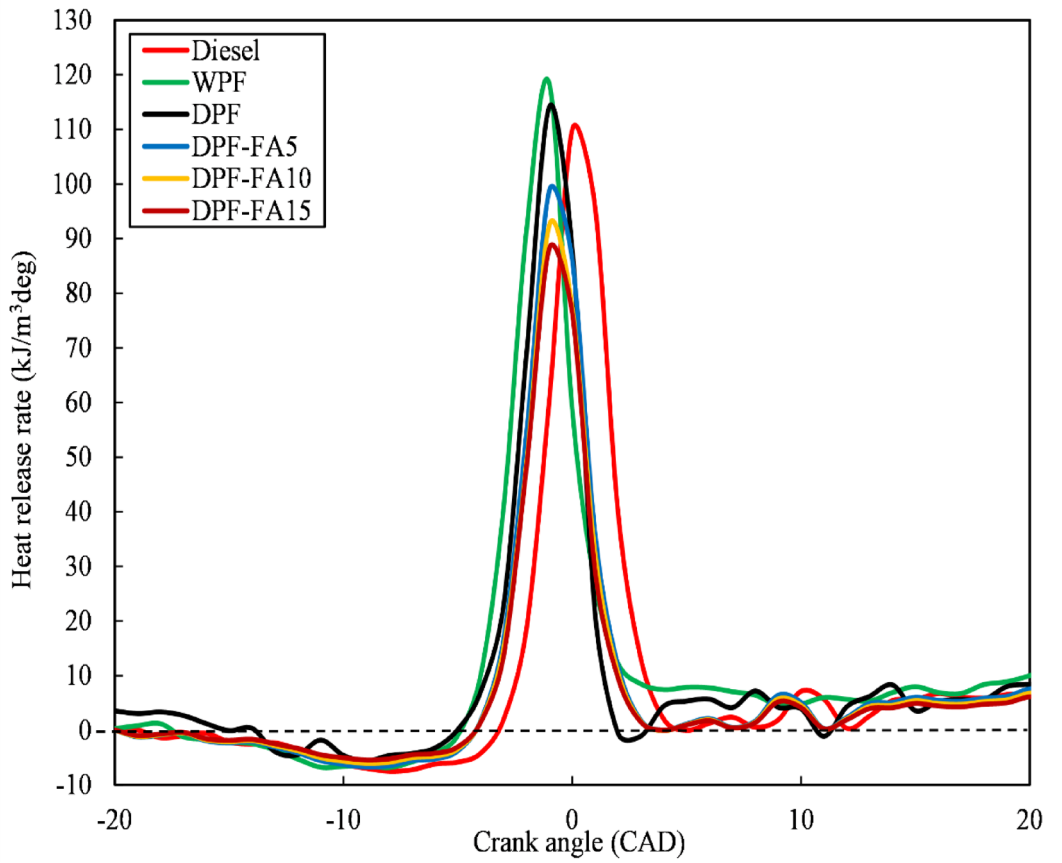


Fig. 5. The variations in heat release rate of test fuels at 100% load.

release earlier in the cycle. The DPF-FA15 reaches its peak CHRR approximately + 9° CAD and diesel at + 12° CAD. As compared with WPF and PDF, improved thermal efficiency could result for DPF-FA15 as enables more work to occur near the top dead center position.

Ignition delay

Ignition delay is the interval between the initiation of ignition and the commencement of main fuel injection. This delay consists of two stages: physical and chemical delay. In physical delay, high-pressure injected fuel droplets are atomized, evaporated, and blended with the existing charge. In chemical delay, chemical reactions commence, and atomized fuel reaches self-ignition, thereby initiating combustion and elevating pressure[34]. The ignition delay period details for all test fuels with an increase in the load are presented in Fig. 7. The period of ignition delay reduces as engine load increases for all fuels. This is due to the elevated in-cylinder temperature and pressure with increased loads, which expedite the chemical reactions during ignition[74]. WPF exhibits the highest ignition delay among all fuels. This results from its low cetane number, higher viscosity, which impede fuel atomization and delays pre-combustion processes[50]. The DPF has a greater ignition delay compared to Diesel, indicating the impact of WPF in the mixture[61]. The ignition delay is shorter than that of pure WPF, suggesting that the combination of WPF with Diesel alleviates certain disadvantages of WPF by enhancing volatility and igniting properties. The enhancement in ignition delay with DPF to WPF underscores the advantages of blending for optimizing combustion characteristics.

Addition of FA enhances the physical characteristics of the DPF blend by decreasing viscosity and increasing volatility. This results in improved atomization and rapid evaporation relative to pure WPF or DPF. With an increase in the amount of FA as in DPF-FA10 and DPF-FA15, the physical delay diminishes due to FA's high volatility, which facilitates the faster dispersion of fuel droplets. In DPF-FA blends, the chemical delay is markedly reduced compared to pure WPF or DPF, as FA accelerates the reaction rates, especially under low to moderate loads. The chemical structure of FA, characterized by oxygenated molecules such as furan, facilitates the generation of essential radicals such as OH•, HO2•. These radicals facilitate the combustion process and promote the shift from the ignition delay phase to the rapid combustion phase. With the increase in FA concentration, the presence of radicals further diminishes the chemical delay, with DPF-FA15 exhibiting the shortest chemical delay among the evaluated blends.

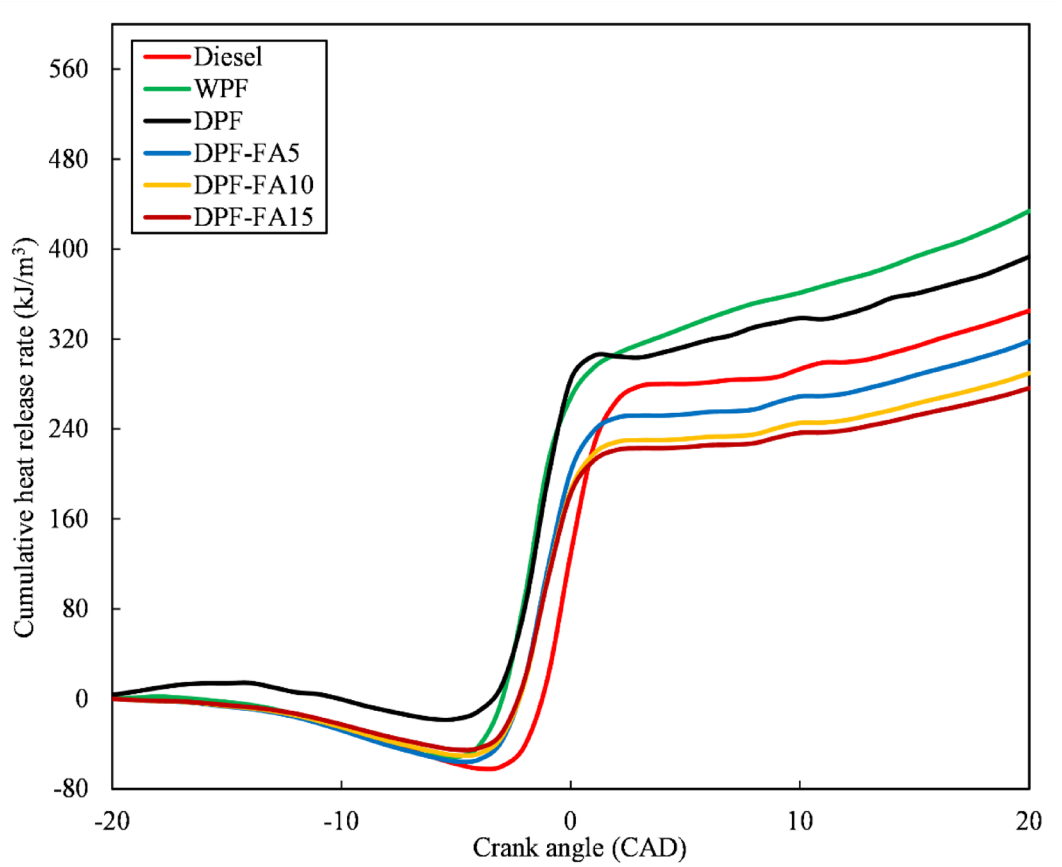


Fig. 6. The variations in cumulative heat release rate of test fuels at 100% load.

Exhaust gas temperature

Engine exhaust gas temperature (EGT) is a metric for the temperature of exhaust gases measured at the exhaust manifold. The EGT reflects the efficient utilization of a fuel's thermal energy. The amount of heat energy converted from fuel to work is decreased by heat loss from the exhaust pipe or an increase in exhaust temperature[34]. The variations of EGT with respect to load is shown in Fig. 8. Under low load conditions, the marginal rise in EGT for DPF-FA blends is ascribed to the slower rates of combustion of WPF constituents in the mixture, necessitating longer time and energy for thorough oxidation, especially at reduced in-cylinder temperatures and pressures. EGT increases with engine load across all fuel types. Under full load conditions, the amount of fuel injected is increased to fulfill power requirements, resulting in enhanced fuel combustion, which increases the in-cylinder temperature along with EGT[46]. At higher load, the EGT for diesel, WPF, DPF, DPF-FA5, DPF-FA10, and DPF-FA15 was found to be 485, 594, 540, 525, 517, and 506 °C, respectively. WPF exhibits the highest EGT of all fuels, attributable to incomplete combustion, because of its higher aromatic content, and prolonged heat release, resulting in extended combustion duration[75]. The DPF exhibits a higher EGT than Diesel, although lower than WPF under all loading conditions. As compared with diesel, EGT for DPF-FA5, DPF-FA10, and DPF-FA15 is higher by 40, 32, and 21 °C, respectively. Enhanced combustion efficiency with increased FA concentration leads to a smooth and cleaner combustion process, hence minimizing heat loss along with exhaust temperatures.

Brake thermal efficiency

Brake thermal efficiency is a vital performance metric that reflects the efficiency of transforming fuel energy into useful output. The variations of BTE for all test fuels are presented in Fig. 9. It is observed that BTE increases with an increase in the load. This results from improved combustion under elevated loads due to increase in-cylinder temperature and pressure, which enhance energy conversion[24]. WPF exhibits the lowest BTE among the evaluated fuels, principally attributable to its inferior ignition characteristics, higher viscosity (Table 2), and decreased cetane number, leading to incomplete combustion along with energy losses[30]. DPF exhibits 12.67% and 12.78% decrease in BTE as compared to diesel at low and high loads, respectively, suggesting that the addition of WPF with diesel partially mitigates the disadvantages of pure WPF, which possesses a lower calorific value and inferior combustion properties[43]. With an increase in the FA content, the BTE of the blends increases. At 100% load, the BTE of DPF-FA5, DPF-FA10, and DPF-FA15 are lower by 10.43, 7.09, and 5.59%, respectively. On the other hand, as compared with DPF, the BTE of DPF-FA5, DPF-FA10, and DPF-FA15 are higher by 2.62, 6.11, and 7.61%, respectively. The availability of additional oxygen from FA promotes low temperature oxidation and reduces the ignition delay which allows a larger fraction of the fuel to burn during the premixed

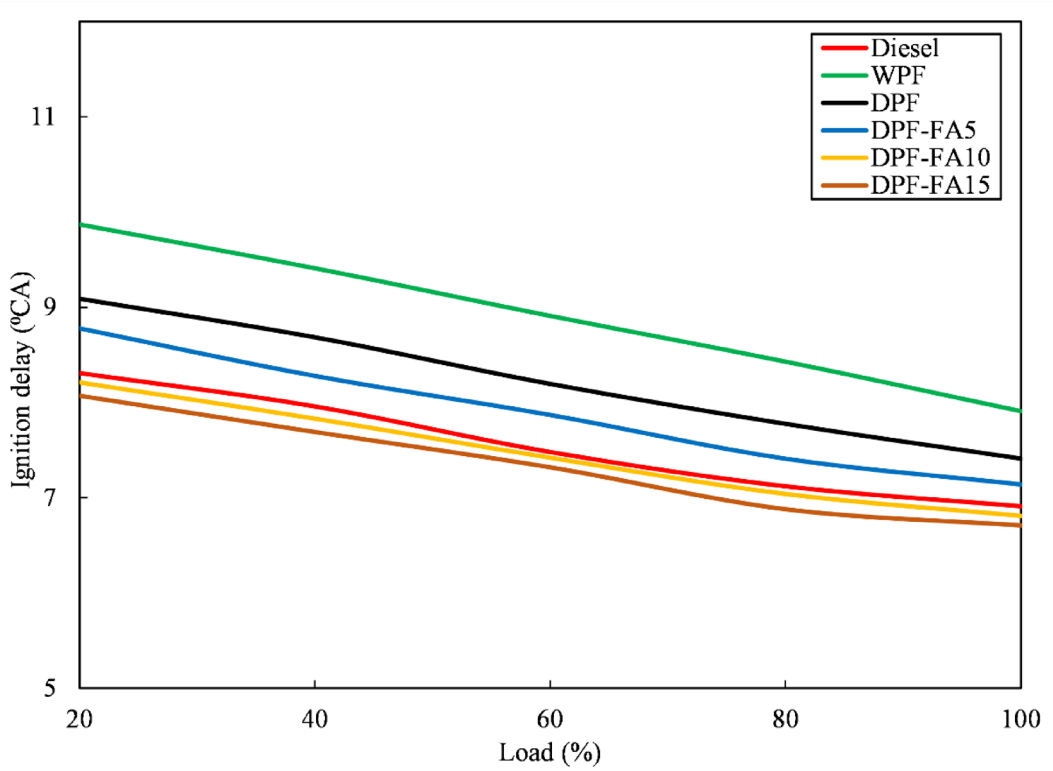


Fig. 7. The ignition delay period details of test fuels.

phase. This results in a more uniform energy release which reduces heat losses and enhances thermal efficiency. FA undergoes heat decomposition into smaller, extremely reactive intermediates including furan, aldehydes, and ketones. These intermediates engage in radical-driven HC chain reactions that expedite the degradation of hydrocarbons in WPF. Radicals such as $OH^{\cdot}$ and $O_2^{\cdot}$ oxidize fuel radicals, resulting in the formation of stable combustion products (CO_2 and H_2O) and the release of energy. The advanced radical pool enhances the combustion efficiency of heavier hydrocarbons in WPF, leading to increased energy conversion and improved BTE.

Brake specific fuel consumption

Brake specific fuel consumption is a critical engine characteristic, defined as the fuel consumption per unit of power during a specified time interval[49]. The variations in BSFC for all test fuels is presented in Fig. 10. With an increase in the load, BSFC for all test fuels reduces due to energy losses at peak load resulting from rapid combustion rates and rapid fuel–air mixing and utilization[35]. WPF exhibits the highest BSFC values, 0.6715 kg/kWh at low and 0.4077 kg/kWh at high loads, among all fuels because of its reduced cetane number, higher viscosity, and lower calorific value, which lead to incomplete combustion and increased fuel consumption[38]. DPF demonstrates, 2.46% at low and 16.5% at high load, higher BSFC relative to Diesel, signifying inferior combustion efficiency attributable to the inclusion of WPF in the blend[61]. However, it exhibits considerable enhancements compared to pure WPF, indicating that integration with diesel minimizes certain limitations of WPF. At 100% load, BSFC for DPF-FA5, DPF-FA10, and DPF-FA15 are higher by 14.73, 12.09, and 9%, respectively as compared to diesel. FA, being an oxygenated additive, enhances the oxidation of heavier hydrocarbons in WPF, leading to stable combustion and less fuel consumption. The improved energy release provided by FA ensures that a larger percentage of the fuel's calorific value is employed for power generation, hence decreasing BSFC. Conversely, the use of FA lowers the viscosity of DPF mixes which results in improved atomization of the fuel spray, optimizing fuel–air mixing and decreasing the time needed for fuel preparation. Enhanced atomization minimizes heat losses and supports combustion, thereby reducing BSFC. Increased evaporation ensures that a larger fraction of the fuel participates in the premixed combustion phase, resulting in reduced BSFC.

Oxides of nitrogen

The thermal mechanism predominantly generates oxides of nitrogen at higher temperatures in diesel engines and, on the other hand, to a lesser extent, via the prompt mechanism. Nitric oxide and nitrogen dioxide react to produce NO_x, with nitric oxide being the primary constituent. Mollo et al.[76] describe the general mechanisms of NO_x production in terms of reactions (6)–(8):

$$O + N_2 \leftrightarrow NO + N \tag{6}$$

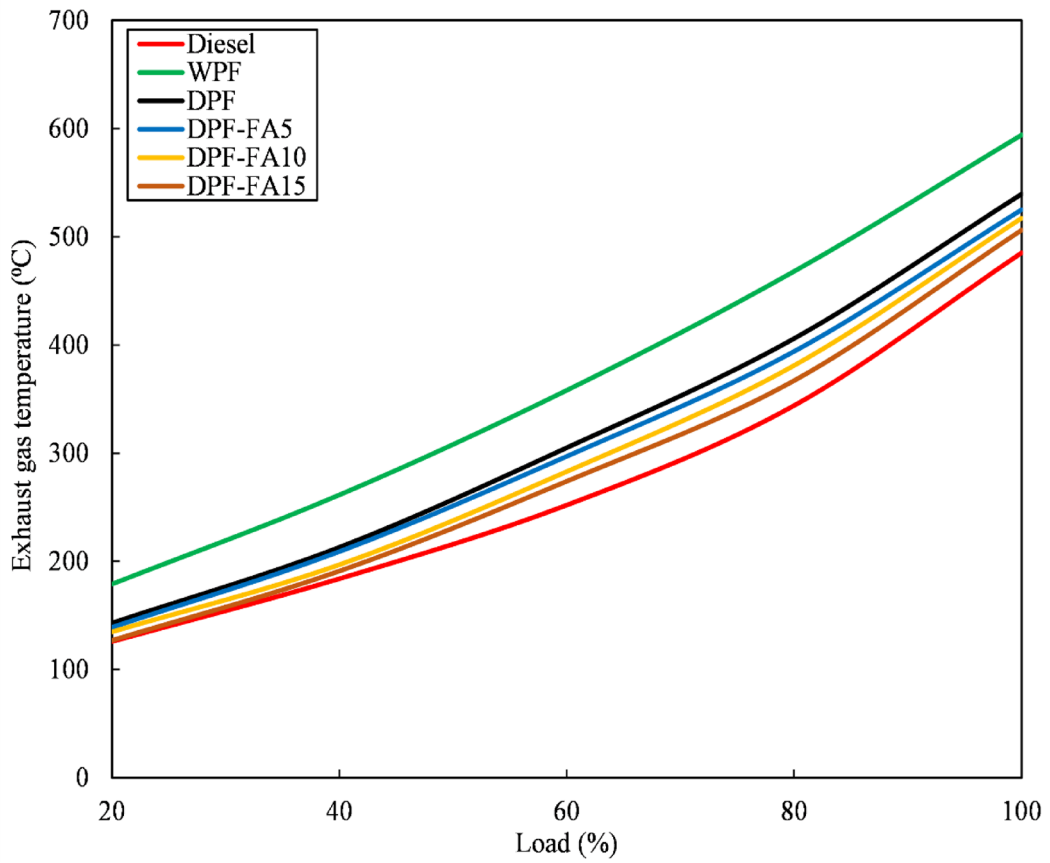


Fig. 8. The variations in exhaust gas temperature of test fuels.

$$N + O_2 \leftrightarrow NO + O \quad (7)$$

$$N + OH \leftrightarrow NO + H \quad (8)$$

Figure 11 illustrates the variance of NO_x for all test fuels with respect to load. At lower load of 20%, low NO_x is reported across all fuels which is attributed to the lower combustion temperature and pressure. However, with an increase in load from 20 to 80%, NO_x also increases exponentially due to the elevated temperatures generated during combustion within engine. This finding is in a good agreement with the finding of Arani et al.[49] and Kaewbuddee et al.[50] who established that inadequate combustion stability resulted in localized temperature increases. At 100% load, the NO_x emission for DPF-FA5, DPF-FA10, and DPF-FA15 are higher by 17.02%, 15.06%, 13.16% and 10.86%, respectively as compared to diesel. According to Januszewicz et al.[52], larger loads lead to an elevated fuel–air ratio, which raises the gas temperature in the combustion chamber, subsequently, the NO_x levels increase. The addition of FA enhances the oxygen availability in the combustion chamber which improves hydrocarbon oxidation but may also elevate thermal NO_x generation by raising localized combustion temperatures [76]. The regulated heat release from FA addition mitigates excessive temperature increases, hence decreasing thermal NO_x generation relative to DPF and WPF. However, the increase in NO_x emissions is less desirable as it significantly contribute to global emissions of air pollutants, which is as critical as climate issues although the increase is within the acceptable range required target[76].

Hydrocarbon

Hydrocarbon emissions mostly arise from insufficient mixing of fuel and air particles during combustion, together with incomplete fuel combustion[30]. The variations of HC emission with respect to load for all test fuels is presented in Fig. 12. HC emissions decrease with increase in the load for all test fuels. The higher in-cylinder temperature and pressure at higher load promote complete combustion which reduces the release of unburned hydrocarbons. Diesel exhibits the lowest HC emissions of all loads due to its high cetane number, and better mixing of fuel with air[34]. WPF exhibits the highest HC emissions among all the tested fuels under low loads. These findings are in good agreement with Tushar et al.[35] in investigating the effect of waste plastic oil on a CRDi engine who established that the poor ignition quality, higher viscosity, and incomplete combustion resulted in highest HC emissions because of its heavier hydrocarbon fractions. DPF exhibits higher HC emissions than diesel across all loads due to the blending of WPF which has a lower cetane number and poor atomization properties. DPF demonstrates significant improvements in HC emissions over neat WPF. From Fig. 11, it is also observed that HC emissions for WPF and DPF are higher than diesel by 20% and 13.6%, respectively, under

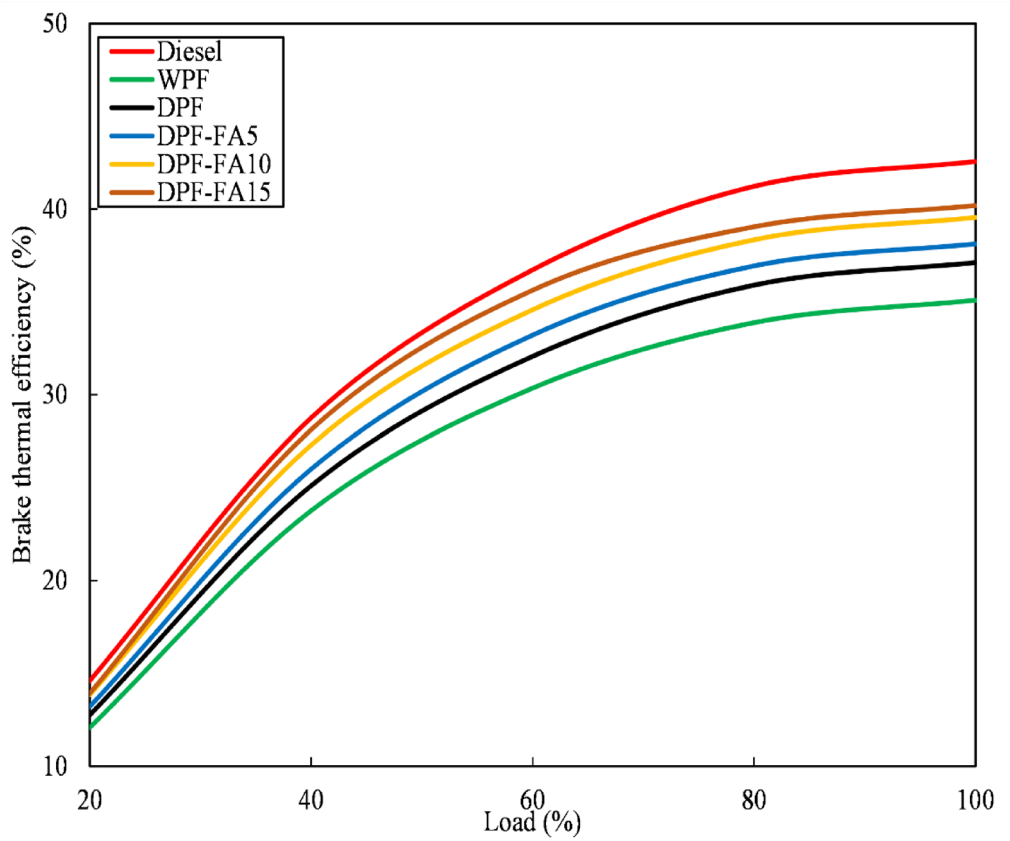


Fig. 9. The variations in brake thermal efficiency of test fuels.

low load conditions and 51.4% and 35.29%, respectively, under full load conditions. According to Pappula et al.[61], the fuel injection quantity increases at high engine loads to meet power demands that lead to rich air–fuel mixtures. The presence of heavier hydrocarbons and lower cetane number in WPF and DPF results in incomplete oxidation in the fuel-rich zones which resulted in higher HC emissions as some hydrocarbons escape unburned. On the other hand, the increased combustion temperature and pressure may not be sufficient at higher loads to fully compensate for the slower ignition and evaporation characteristics of WPF and DPF. HC emissions for DPF-FA5, DPF-FA10, and DPF-FA15 are higher than diesel by 10.74%, 9.24%, and 6.89%, respectively. This is attributed to the lower calorific value of FA which requires more energy for complete combustion under low load conditions. At 100% load, HC emissions for DPF-FA5, DPF-FA10, and DPF-FA15 are higher than diesel by 31.25%, 23.25%, and 13.15%, respectively. FA provides additional oxygen during combustion which promotes the oxidation of hydrocarbons and reduces unburned fuel particles. The addition of FA enhances combustion stability which reduces ignition delay and promotes faster pre-combustion reactions. This leads to more efficient energy release and reduces the emission of unburned hydrocarbons at higher FA concentrations.

Carbon monoxide

Carbon monoxide (CO) emission mostly arise from insufficient oxygen supply or inefficient oxygen utilization during combustion. The CO emission for test fuels under various loads are illustrated in Fig. 13. In general, diesel engine operates at lean mixtures which results in reduced CO emissions. It can be seen in Fig. 12 that CO emission in diesel engine decreased progressively from 20% load to 80% load and then increased exponentially up to 100% load for all test fuels. The CO emission trend in Fig. 13 is governed by load dependent combustion efficiency, mixing of fuel and air mixing and availability of oxygen. In diesel engines, combustion occurs under lean conditions at lower loads, such that an excess of air is available for oxidation that leads to better fuel combustion and lower CO emissions. An increase in load from 20 to 80% elevates the in-cylinder temperature, hence improving fuel atomization, mixing, and oxidation, which subsequently lowers CO production[43]. CO mostly arises from incomplete combustion, and under moderate loads, the air–fuel ratio and in-cylinder temperature are sufficient, hence reducing CO emissions. At 100% load, the fuel injection rate increases significantly to meet power demand that leads to local fuel-rich zones inside the combustion chamber. High fuel concentration in certain regions results in insufficient oxygen availability that leads to partial oxidation of hydrocarbons and a sharp increase in CO emissions[46]. In addition to that, higher combustion temperatures at full load cause reduced air entrainment that further inhibit complete oxidation. A similar trend has been reported in previous studies where CO emissions rise at full load due to oxygen starvation and localized rich combustion pockets[24]. The addition of FA partially compensates for this by enhancing fuel–air mixing, but at 100% load, oxygen deficiency

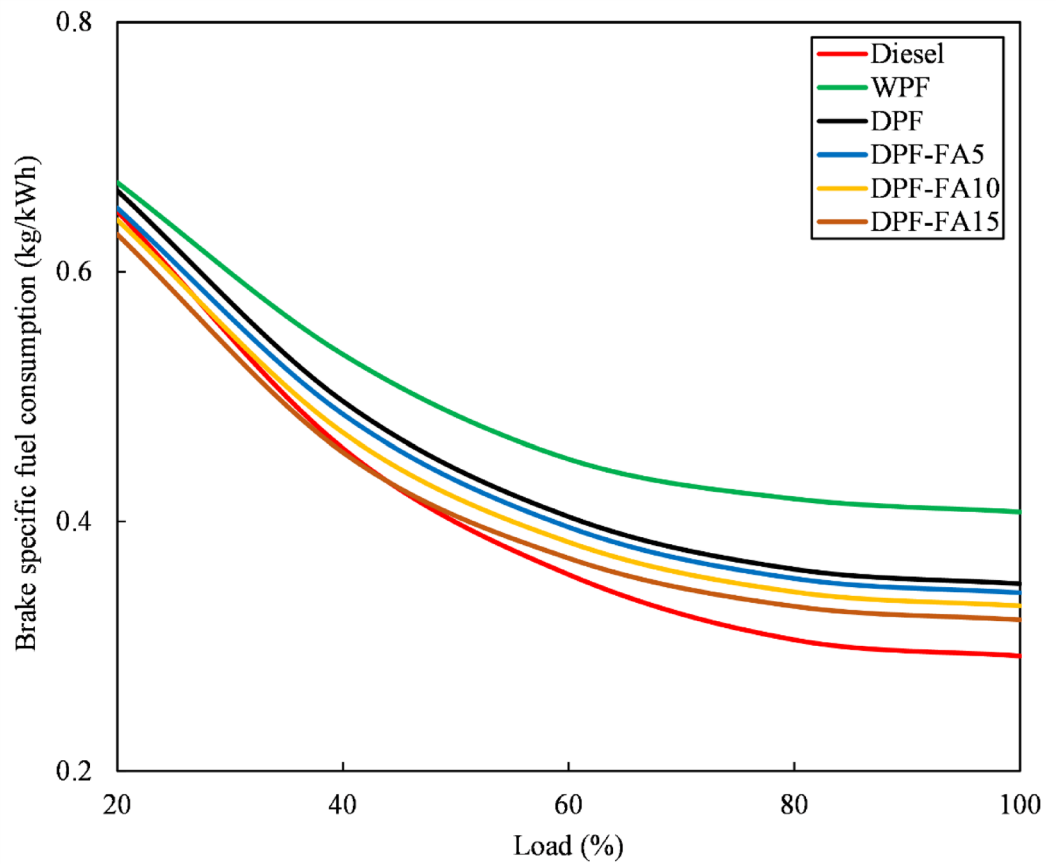


Fig. 10. The variations in brake specific fuel consumption of test fuels.

still dominates that lead to exponential CO rise. Diesel consistently exhibits the lowest CO emission due to its high cetane number, and optimized combustion efficiency Desai et al.[38].

WPF produces the highest CO emission at low and high loads. As compared with diesel, CO emission for WPF is higher than diesel by 6.99% and 23.59%, respectively, at low and high load conditions. This is attributed to the poor ignition properties, lower cetane number, and increased viscosity which results in inefficient atomization of the fuel–air mixture[34,45]. CO emission for DPF is higher than diesel by 3.16% and 13.44%, respectively, at low and high loads but lower than WPF by 3.94% and 11.82% at low and high loads, respectively. These results support the findings of Ellappan and Pappula[43] who established that adding WPF with diesel improves oxidation kinetics and reduces CO emission.

It can also be seen from Fig. 12 that CO emission for DPF-FA5, DPF-FA10, and DPF-FA15 are higher than diesel by 1.64%, 1.5%, and 1.03%, respectively at low load and 11.06%, 7.37%, and 3.74%, respectively, at high load. This can be attributed that FA reduces the viscosity of the fuel blend which leads to better atomization and air fuel mixing. This results in a more homogenous charge and more complete combustion and CO reduction. According to Padmanabhan et al.[30], the addition of alcohol with low cetane numbers prolongs the ignition delay period in combustion processes, which results in reduced CO emission. Rajesh kumar[77] reported that the predominant effect of the OH group leads to abstraction of hydrogen atoms by hydroxyl radicals (OH•) radicals from the carbon position in most alcohols. FA contributes chemically bound oxygen that actively participates in the combustion process. FA undergoes thermal decomposition during the ignition delay period and produces reactive intermediates such as furans, aldehydes, and OH•. These intermediates accelerate low temperature oxidation and facilitate the formation of hydroperoxides (ROOH) that are critical for propagating chain reactions in combustion.

Carbon dioxide

Carbon dioxide (CO_2) emissions are a direct indicator of complete combustion. Higher CO_2 emissions typically imply better oxidation of hydrocarbons and lower emissions of unburned fuel such as CO and HC[61]. The CO_2 emission trends for test fuels, as illustrated in Fig. 14, reflect the combustion quality of these fuels under different engine loads. CO_2 emissions increase with load for all tested fuels as higher loads demand more fuel to generate power which lead to increased fuel combustion resulting in higher CO_2 emissions[78]. At 20% load, the differences in CO_2 emissions between the fuels are less pronounced due to lower combustion temperatures and pressures. As compared with diesel, WPF and DPF emit 3.54%, 1.77%, respectively, lower than Diesel and this is attributed to the poor combustion characteristics of WPF which include lower cetane number and incomplete oxidation

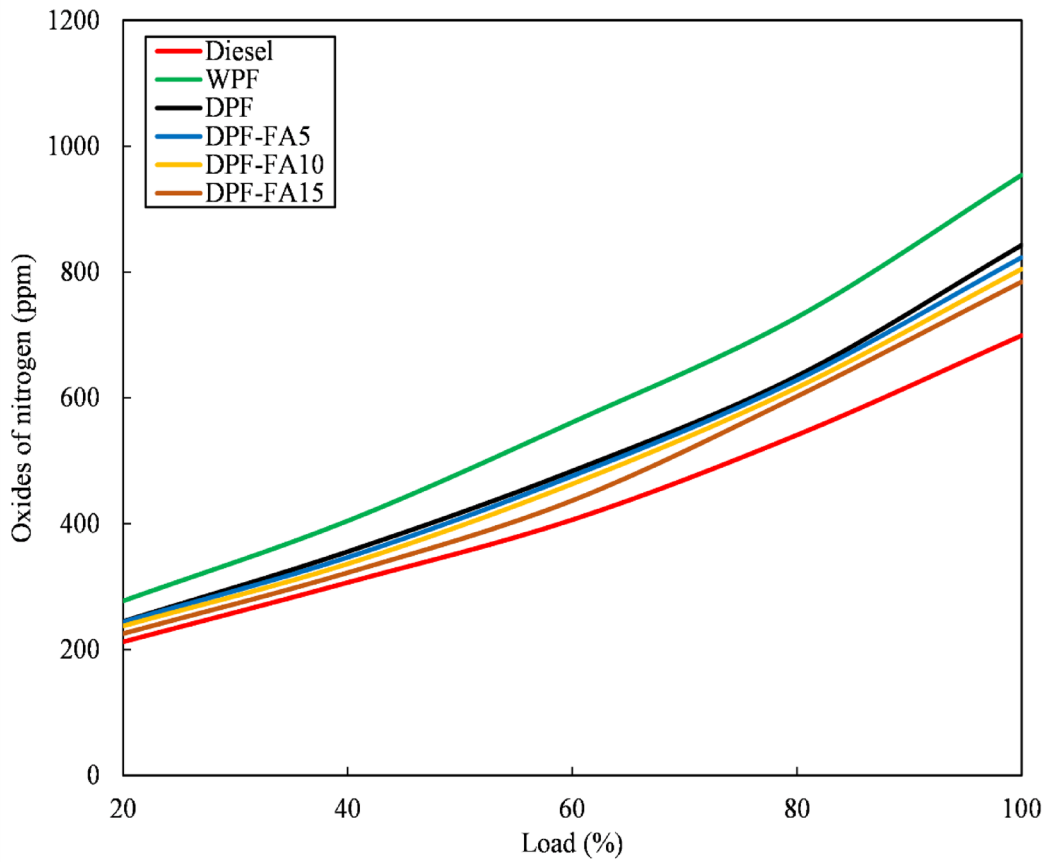


Fig. 11. The variations in oxides of nitrogen for test fuels.

of heavier hydrocarbons[79]. CO_2 emissions for WPF and DPF at 100% load were found to be lower than diesel by 35.24% and 18.87%, respectively. Güngör et al.[80] also reported that higher fuel consumption, in-cylinder temperature, and pressure promote better combustion which results in more pronounced CO_2 emissions. The addition of FA to DPF blends has a pronounced impact on CO_2 emissions as FA provides additional oxygen during combustion that facilitate low temperature oxidation of HC. This results in higher CO_2 emissions which indicate better combustion. As compared with diesel, CO_2 emissions for DPF-FA5, DPF-FA10, and DPF-FA10 were found to be higher by 5.42%, 7.64%, and 9.41%, respectively at low load and 5.39%, 7.62%, and 19.47%, respectively at high load. FA reduces ignition delay period and improves combustion phasing which ensures complete oxidation of HC during the combustion process. The higher FA concentrations in DPF-FA10 and DPF-FA15 lead to significant increase in CO_2 emissions that reflects in improved energy conversion and less unburned fuel. Low temperature oxidation of FA promotes reduction in CO and HC emissions while shifting the carbon balance toward CO_2.

Smoke

One of the visible pieces of evidence of combustion process comes from the engine exhaust. Inefficient fuel combustion that leaves unburned carbon particles generates smoke, which is usually formed in the diffusion combustion phase[30]. Atomized fuel droplets break down into carbon atoms and are then oxidized in the combustion zone[31]. Limited air availability, a high carbon-to-hydrogen ratio, higher fuel viscosity, poor atomization, and too high fuel accumulation in the combustion chamber cause smoke emissions in fuel rich zones[63]. Smoke opacity increases with load for all tested fuels, as shown in Fig. 15, as higher loads require more fuel injection which leads to higher fuel air equivalence ratios and localized fuel rich zones[64]. WPF shows the highest smoke opacity among all fuels that reflects its poor combustion quality, higher aromatic content, and soot precursor formation[24,46,49,61]. Smoke opacity differences are less pronounced at 20% load due to relatively lower fuel injection and moderate in-cylinder temperature. Smoke emissions for WPF and DPF are higher than diesel by 29.58% and 15.15%, respectively. The differences in smoke opacity between the fuels at 100% load are more pronounced due to increased fuel injection and higher equivalence ratios which lead to more significant variations in soot formation[34,45]. Smoke emissions for WPF and DPF are higher than diesel by 28.07% and 14.04%, respectively. The addition of FA to DPF blends significantly reduces smoke opacity by improving combustion efficiency and suppressing soot precursor formation. The reduction in smoke opacity with FA addition, at low load, is moderate due to lower in-cylinder temperature and pressure[81]. As compared with diesel, smoke emission at 20% load for DPF-FA5, DPF-FA10, and DPF-FA15 is found to be higher by 14.67%, 11.76%,

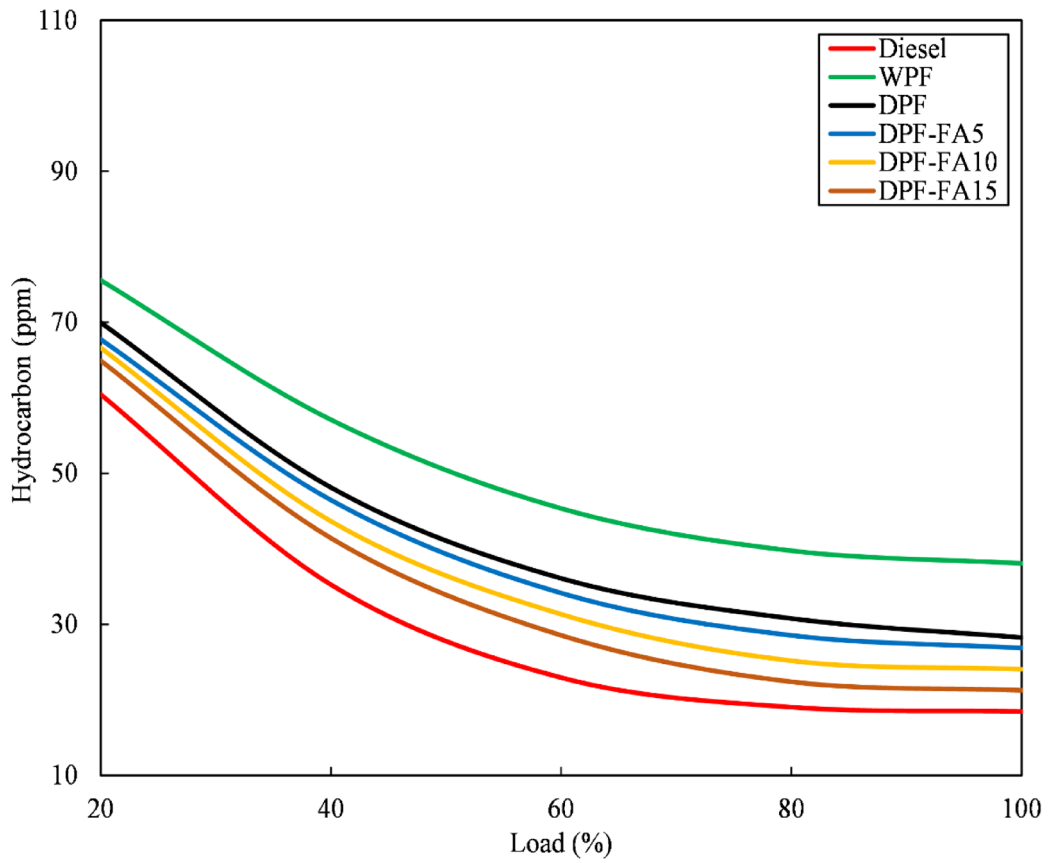


Fig. 12. The variations in hydrocarbon emission for test fuels.

and 8.70%, respectively, and at 100% load, it is higher by 12.80%, 11.12%, and 8.45%, respectively. FA introduces chemically bound oxygen that enhances oxidation of HCs and reduces the formation of polycyclic aromatic hydrocarbons which are key precursors to soot. Under high load conditions, FA reduces ignition delay period and enhances premixed combustion that results in more efficient combustion and reduced soot formation[82,83]. The decomposition intermediates of FA delay the formation of carbonaceous particles and promote the oxidation of carbon into CO and CO_2 instead of contributing to smoke emission.

Optimization results of test fuels

The optimization results of the PI calculation, as shown in Table 5, are used to compare the performance of DPF and its blends with FA (DPF-FA5, DPF-FA10, and DPF-FA15) under various engine loads with maximum Kissing Number of 6. The results focus on the interdependencies between combustion efficiency, fuel consumption and emission control under optimal configurations for each load.

The selection of optimal blend using PI based on optimization using Kissing Numbers adheres to the following specific criteria to ensure systematic and unbiased identification of the most suitable configurations:

- The PI for each configuration is required to be unique between the optimal combinations. Those configurations that have the same PI value at different load conditions are excluded to reduce the redundancy in the optimization.
- In the selection process, only the configurations with negative PI values are considered. Negative PI values indicate suboptimal trade-offs in performance and emissions that align with the goal of identifying configurations for improvement.
- The configuration with the highest negative PI is selected as the optimal combination from among the eligible configurations. This ensures that the configuration balances performance and emissions most effectively under the given load condition.
- The optimal configuration for each load condition must be unique to avoid overlapping between configurations to ensure distinct recommendations for every load scenario.

Based on the above criteria, DPF-FA15 at 80% load is found to be an optimal combination. The uniform PI values across configurations indicate that the benefits of FA at high loads are consistent. The better performance of DPF-FA15 can be attributed to the higher oxygen content in FA that promotes complete combustion and reduces the formation of soot precursors and improve fuel efficiency and emissions control. In the investigation on performance and emissions on plastic pyrolysis oil, Afzal et al.[84] observed that the diesel blend of 20%

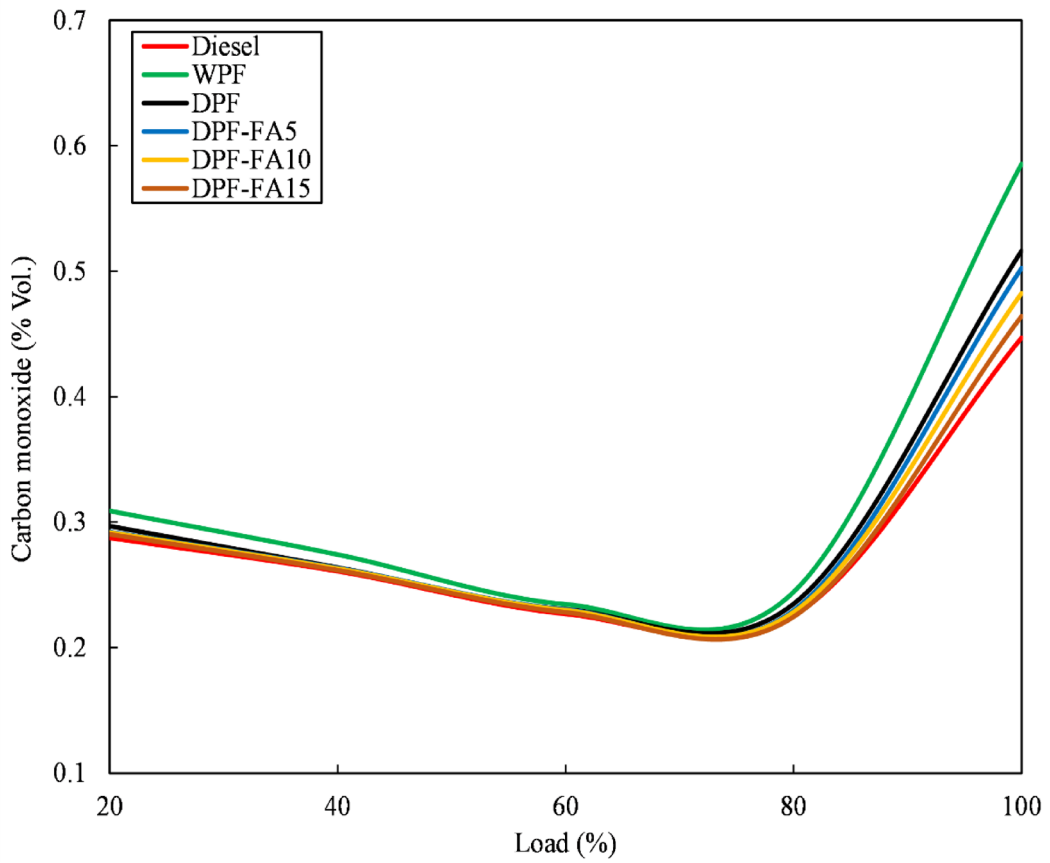


Fig. 13. The variations in carbon monoxide emission for test fuels.

plastic oil exhibited better performance at 75% load adapting RSM. In another investigation to find the optimal combination of engine operating parameters using RSM and ANN, Uslu[85] found that 17.88% of palm oil blend, 35° crank angle and 0.78 kW engine load was optimal.

Validation of optimization using Kissing Numbers with Grey relational analysis

Grey Relational Analysis (GRA) is adapted in the present study to validate the results of Kissing Numbers optimization. GRA is a multi-criteria decision-making method based on grey system theory that is widely used to analyze complex relationships among multiple dependent parameters[86]. GRA provides a quantitative measure of the closeness of a set of experimental results to an ideal reference.

To eliminate the influence of different units and magnitudes among the performance and emission parameters, data normalization was performed as detailed in Section "Data normalization". After normalization, the Grey Relational Coefficient (GRC) was computed to determine the closeness of each experimental result to the ideal performance using Eq. (9)[87].

$$GRC_{ij} = \frac{\Delta_{\min} + \zeta\Delta_{\max}}{\Delta_{ij} + \zeta\Delta_{\max}} \tag{9}$$

where, $\Delta_{ij} = \left|X_{0j}^{*} - X_{ij}^{*}\right|$ is the absolute difference between the optimal or reference value and the normalized experimental value, $\Delta_{\min}$ and $\Delta_{\max}$ are the minimum and maximum differences among all experimental values, and ζ is the coefficient to maintain a balanced weightage ($\zeta = 0.5$).

The Grey Relational Grade (GRG) was computed by averaging the GRC across all performance and emission parameters, Eq. (10), which provides a single index to rank the fuel blends[88].

$$GRG_i = \frac{1}{n}\sum_{j=1}^{n} w_k GRC_{ij} \tag{10}$$

where GRG_i is the GRG for the *i*th experiment, w_k is the normalized weight of the factor $\left(\sum_{k=1}^{m} {}_k = 1\right)$ and n is the total number of parameters.

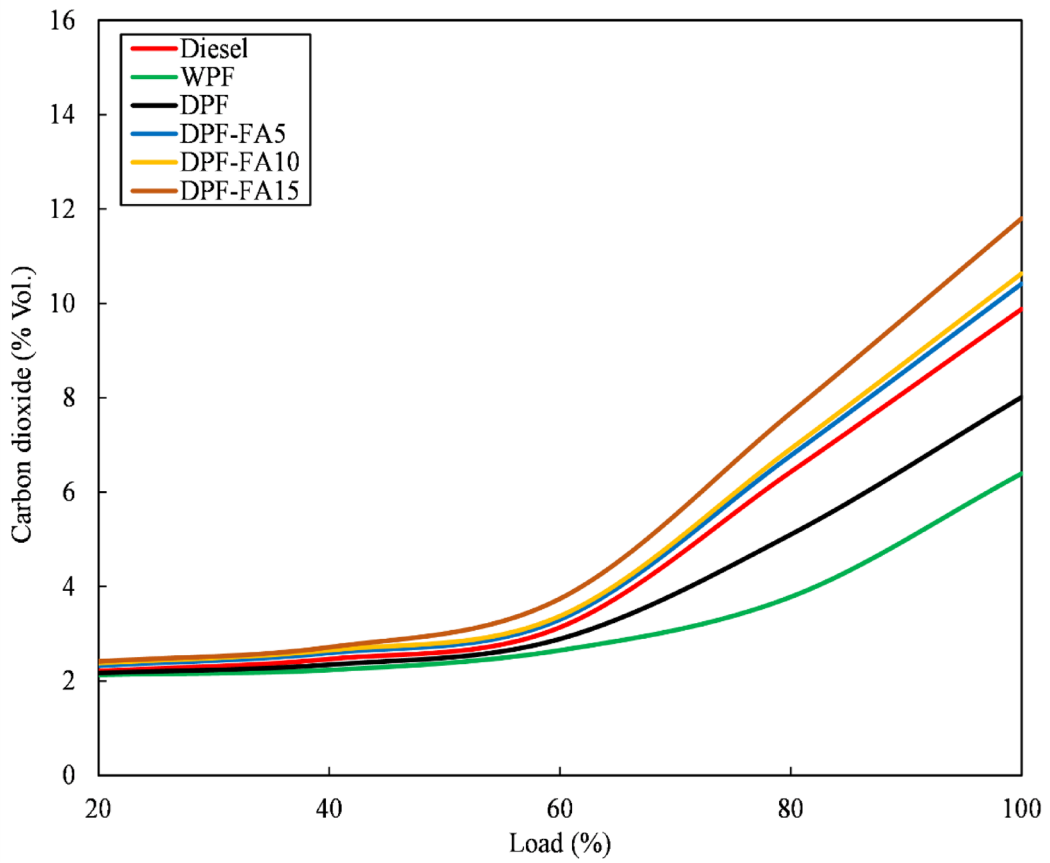


Fig. 14. The variations in carbon dioxide emission for test fuels.

The higher the GRG value, the closer the experimental condition is to the optimal reference, and thus the better its overall performance. The GRG values for the performance and emission characteristics of DPF, DPF-FA5, PDF-FA10, and DPF-FA15 are presented in Table 6.

Table 6 provide a comprehensive evaluation of the performance and emission characteristics of different fuel blends based on the GRG values. Among the tested blends, DPF-FA15 exhibited the highest GRG values (1.000) across all parameters that confirms its superiority in both engine performance and emissions. The BTE increased with FA addition exhibiting increase in GRG from 0.333 for DPF to 1.000 for DPF-FA15 that highlight the role of oxygen enrichment in improving combustion efficiency. Similarly, BSFC decreased that indicate better fuel utilization. NO_x levels were higher with FA because of higher peak combustion temperatures, and this is reflected in the GRG values that rise from 0.333 (DPF) to 1.000 (DPF-FA15). However, this trade-off is counterbalanced by the significant decrease in smoke opacity, HC and CO levels, all of which had progressively higher GRG values of 1.000 for DPF-FA15. This trend suggests that FA enhances combustion thereby leading to lower emissions of unburned fuel. Besides that, the increase in CO_2 emissions indicates that more complete carbon oxidation leads to higher CO_2 emission, which is a good indication of complete combustion and lower of harmful pollutants such as CO and HC[89].

The GRA results validate the Kissing Numbers optimization and confirms that DPF-FA15 is the optimal blend for achieving a balance between fuel efficiency and emissions control and reinforce the reliability of the proposed optimization approach. This highlights the potential of FA as a promising oxygenated additive for improving DPF blend combustion. Despite the increase in NO_x emissions, this issue can be mitigated using exhaust gas recirculation or after-treatment systems that makes DPF-FA15 a highly suitable candidate for practical engine applications.

Merits of optimization using Kissing Numbers

Optimization using Kissing Numbers method offers several advantages over existing optimization techniques in analyzing and optimizing the performance of alternative fuel blends. These are based on the capability to handle multi-dimensional data and its ability to offer solutions at given operating conditions of engines. The following are the key benefits:

- As for traditional methods like RSM, the optimization using Kissing Numbers incorporates several performance characteristics into a single PI. This ensures that the optimization process considers all the objectives simultaneously and in an unbiased manner.

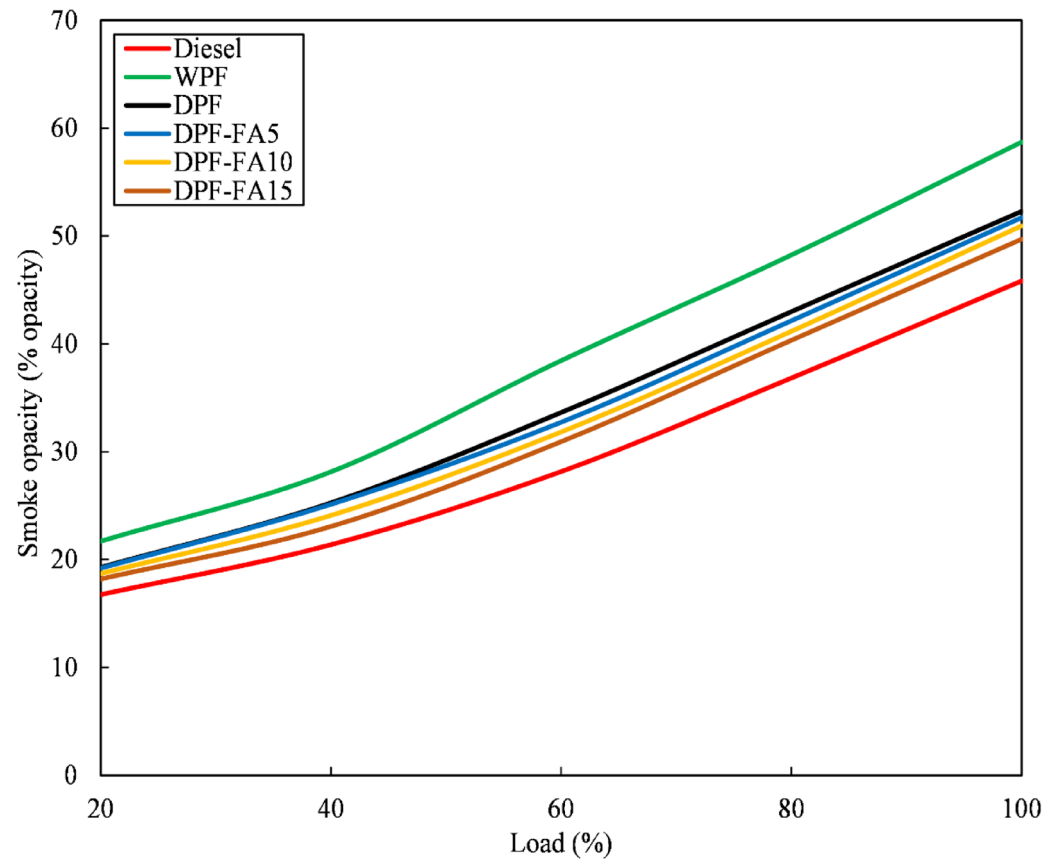


Fig. 15. The variations in smoke emission for test fuels.

S. no	Load	PI			
		DPF	DPF-FA5	DPF-FA10	DPF-FA15
1	20	-0.302	-0.302	-0.302	-0.302
2	40	-0.200	-0.200	-0.200	-0.201
3	60	-0.116	-0.118	-0.115	-0.114
4	80	-0.001	-0.001	-0.001	-0.001
5	100	0.100	0.100	0.100	0.100

Table 5. The results of optimization using Kissing Numbers at various loads.

Blend	BTE	BSFC	EGT	NO_x	Smoke	HC	CO	CO_2
DPF	0.333	0.333	0.333	0.333	0.333	0.333	0.333	0.333
DPF-FA5	0.429	0.411	0.414	0.387	0.387	0.413	0.460	0.569
DPF-FA10	0.713	0.569	0.593	0.514	0.550	0.586	0.549	0.655
DPF-FA15	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000

Table 6. The GRG values for performance and emission characteristics.

- Optimization using Kissing Numbers is a heuristic approach motivated by geometric and mathematical ideas, i.e., Kissing Numbers, unlike computationally intensive techniques like ANN and GA. This keeps precision in identifying ideal solutions while lowering the computational load.

- The symmetry of optimization using Kissing Numbers makes it easy to understand the relationships between configurations and load conditions and therefore visualize them. This feature is particularly useful for understanding the trade-offs between fuel efficiency, combustion efficiency and emissions control.
- Methods like RSM and ANN depend heavily on precise experimental data for calibration and GA may encounter convergence issues in high dimensional spaces. Optimization using Kissing Numbers avoids these pitfalls by focusing on spatial packing efficiency and iterative refinement that ensure robust and reliable outcomes.

Limitations of the present study

Optimization using the Kissing Numbers approach has considerable benefits in simplifying and enhancing the optimization process, although it features inherent limits. These limitations underscore aspects where the technique may inadequately address dynamic situations. The key limitations are as follows:

- Optimization using Kissing Numbers relies on geometric principles and heuristic approaches that may oversimplify complex engine dynamics such as non-linear interactions between combustion parameters and emissions.
- The optimization process focused on a fixed set of parameters such as performance and emissions. Other potentially influential factors such as combustion stability or particulate matter size distribution were not incorporated that could affect the comprehensiveness of the results.

Conclusion

This study investigated the potential of furfuryl alcohol as an oxygenated additive to improve the performance and emission characteristics of diesel plastic fuel blends in a common rail direct injection engine. An extensive experimental investigation was performed to assess the combustion, thermal efficiency, fuel consumption, and emissions of several test fuels (DPF, DPF-FA5, DPF-FA10, DPF-FA15) under varied engine loads. Optimization using Kissing Numbers was applied to identify the optimal configuration for each load condition based on the performance index that integrates key parameters such as brake thermal efficiency, brake specific fuel consumption, and emissions. The findings of this study can be summarized as follows:

- Compared to diesel, DPF-FA blends exhibited lower peak cylinder pressures due to the lower calorific value of FA. For instance, DPF-FA15 showed a 4.3% lower peak cylinder pressure at 80% load compared to diesel.
- The maximum HRR for DPF-FA15 was 6.5% lower than diesel at 100% load, which is attributed to the slower combustion of WPF components. Furthermore, the oxygen content in FA improved combustion stability compared to neat DPF.
- BTE for diesel at 100% load was 42.56%, while DPF-FA15 achieved 40.18%, reflecting a 5.6% decrease compared to diesel. At moderate loads, DPF-FA5 achieved a BTE of 26.00% lower than diesel (28.75%) by 9.6%, which demonstrates the positive impact of FA on DPF efficiency.
- Diesel exhibited the lowest BSFC with 0.292 kg/kWh at 100% load. DPF-FA15, with improved combustion efficiency, exhibited 0.321 kg/kWh, which is a 9.03% increase in fuel consumption compared to diesel. At moderate loads, DPF-FA5 achieved a BSFC of 0.485 kg/kWh, which is 5.8% higher than diesel but significantly better than neat DPF (0.533 kg/kWh).
- CO emissions for diesel at 100% load were 0.45%Vol., whereas DPF-FA15 emitted 0.46%Vol. with a 2.17% reduction. Neat DPF emitted 0.52%Vol., which indicates the beneficial impact of FA in reducing CO emissions.
- CO_2 emissions for diesel at 100% load were 9.88%Vol., whereas DPF-FA15 emitted 11.8%Vol. with a 16.27% reduction.
- Diesel recorded HC emissions of 18.48 ppm at 100% load, while DPF-FA15 emitted 21.28 ppm, which represents a 13.1% increase. This increase was due to the heavier hydrocarbons in WPF, regardless of the oxidation-enhancing characteristics of FA.
- Diesel exhibited 700 ppm NO_x emissions at 100% load. DPF-FA15 emitted 784 ppm, which is a 10.7% increase. This is attributed to higher localized combustion temperatures caused by FA addition.
- The optimization using Kissing Numbers approach identified DPF-FA15 as the most suitable blend for high loads.

This study demonstrates that while diesel maintains better brake thermal efficiency and combustion stability, DPF-FA15 blend offers promising improvements in emissions and competitive performance metrics. These results show the potential of FA as a viable option for use with WPF-diesel blends with a view to solving the plastic waste management problem while remaining environmentally friendly. Future work should explore the long-term durability and after-treatment compatibility of FA blends to enhance their practical viability.

Data availability

The data that support the findings of this study are available on request from the corresponding author.

Appendix A

```
import numpy as np

import pandas as pd

data = {"load": [20, 40, 60, 80, 100]

        "DPF": [12.75, 25.11, 32.07, 35.90, 37.12]

        "DPF-FA5": [13.20, 26.00, 33.21, 36.94, 38.12]

        "DPF-FA10": [13.84, 27.29, 34.57, 38.34, 39.54]

        "DPF-FA15": [13.91, 28.11, 35.62, 39.04, 40.18]}

df = pd.dataframe

configs = ["DPF", "DPF − FA5", "DPF − FA10", "DPF − FA15"]

kissing_results = [ ]

if i < j;

kissing_results.append
{"config1": config1, "config2": config2, "load": min digit_row["load"]}

kissing_dt = p.dataframe(kissing_results)
```

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Declarations

Competing interests

The authors declare no competing interests.

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